
When Does a Small Model Know to Hand Off? Zero-Training Escalation Gates for Full-Duplex Speech Models

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Abstract

1 Full-duplex speech models must answer in real time with a small on-device back-
2 bone, yet many queries exceed that backbone’s competence. We study the first
3 half of a two-stage design: can the small model itself signal, with *zero additional*
4 *training*, that a query should be escalated to a large cloud model? We attach
5 candidate gate signals—decode-time scalars, linear probes on hidden states, and
6 verbalized self-evaluation ($p(\text{True})$)—to MiniCPM-o 4.5 (9B, omni/duplex fine-
7 tune of Qwen3-8B) and evaluate them against judged failure labels on 600 frozen
8 public-benchmark queries, with matched-pair controls across ten backbones and
9 both text and speech input. We find that (i) a standard final-layer probe is largely a
10 query-type shortcut that inverts under distribution shift, while pre-answer $p(\text{True})$
11 catches 100%-fail long-tail traps with no training; (ii) duplex fine-tuning *over-*
12 *writes self-knowledge readouts without destroying the underlying information*:
13 mid-layer probe transfer is near the raw backbone’s (leave-one-pool-out math
14 0.931 at layer 22 vs. 0.366 at layer 35), and the damage is specific to the late-layer
15 last-token position where a streaming head encodes turn control; (iii) the two read-
16 outs fail in complementary modalities—the late-layer cliff is text-input-specific,
17 whereas verbalized introspection collapses on audio input because the verbal self-
18 check runs over text-token pathways that audio embeddings do not feed; (iv) the
19 modality-shared mid-layer core that makes cheap text-side calibration transfer to
20 speech is created by end-to-end training and is absent under a frozen-backbone
21 adapter. As a system, the resulting mid-layer gate decides in 20 ms (text) / 45 ms
22 (audio)—before the first output token—and gated-cloud escalation dominates the
23 model’s own thinking mode on both accuracy and latency at every escalation rate
24 (e.g. 0.787 vs. 0.637 accuracy at $3.4\times$ lower latency at a 33% escalation budget).
25 We close the loop live: in full streaming duplex sessions, an end-of-turn mid-layer
26 read decides before the first output token (AUC .887), the returned expert answer
27 is spoken by the talker—the same low-confidence state that fires the gate predicts
28 relay compliance, so escalation and injection are two uses of one signal—and the
29 live system climbs monotonically from its honest streaming floor (.422 $\rightarrow$.633
30 at 57% escalation on speakable content), with the gap to offline operation decom-
31 posed into measured audio, streaming, and speech-channel taxes. Finally, the gate
32 transfers *frozen*: on six public speech benchmarks it fires at sane rates with only
33 label-free threshold quantiles, lifts Speech TriviaQA to .860 live (official offline
34 anchor: .755; a 30B omni model: .629), matches the cloud expert at half the expert
35 calls on Llama Questions, and the mid-network readability it exploits replicates on

a second, architecturally disjoint duplex family (hybrid-Mamba NemotronLabs-VoiceChat-11B) — evidence that duplex models as a class carry a free, readable self-assessment signal that standard readouts miss.

1 Introduction

Voice assistants built on full-duplex speech models [Défossez et al., 2024, Zeng et al., 2024, Xu et al., 2025] face a structural tension. The conversational loop—listening while speaking, turn-taking inside a $\sim 200\text{--}300$ ms budget [Stivers et al., 2009]—demands a small, low-latency model close to the user. Answer quality on knowledge- and reasoning-heavy queries demands a large one. The natural architecture is a *talker-thinker* split, echoing the System 1/System 2 distinction [Kahneman, 2011]: a small duplex “talker” holds the live session and answers by default, and a large cloud “thinker” is consulted only when needed. This paper is about the load-bearing decision in that architecture: **when should the talker hand off?**

We deliberately restrict ourselves to *zero-training* gate signals — quantities that can be read off the small model as-is, without fine-tuning it: decode-time scalars (entropy, margin), linear probes on hidden states [Alain and Bengio, 2016], and verbalized self-evaluation, i.e. $p(\text{True})$ -style “Would you answer this correctly?” prompts [Kadavath et al., 2022]. The constraint is practical (duplex fine-tunes are expensive and fragile to retrain) but it also sharpens the scientific question: *does a small full-duplex model already know what it does not know, and can that knowledge be read out in time to act on it?*

Our answer, developed over a controlled study of ten backbones/fine-tunes and both input modalities, is: **the knowledge is there; the readouts are what break, and they break in systematic, duplex-fine-tune-induced ways.** Concretely:

1. **Standard probes flatter to deceive (§4).** A final-layer hidden-state probe reaches AUC 0.822, but an audit shows it is mostly a query-*type* recognizer: a pool-identity oracle alone scores 0.715, and leave-one-pool-out (LOPO) transfer *inverts* on math (0.372). Pre-answer $p(\text{True})$ needs no training, beats the probe on most backbones, and scores 100%-fail SimpleQA traps at 0.945 — but only if asked *before* the model answers; after answering, it endorses its own confident-wrong answer ($0.945 \rightarrow 0.360$).
2. **Duplex fine-tuning overwrites readouts, not knowledge (§5).** Across matched pairs (raw vs. omni-streaming vs. duplex fine-tunes of the same backbone), LOPO probe transfer degrades in the order raw > omni-streaming > duplex, with label coverage controlled. A full layer $\times$ position sweep localizes the damage: on MiniCPM-o 4.5, LOPO math is 0.931 at layer 22 but 0.366 at layer 35, with a sharp cliff over the last four layers, only at the last-token position — exactly where a streaming head must encode turn-control state.
3. **Each readout has one blind modality (§6).** The late-layer cliff is text-input-specific: under audio input the final layer does not invert (0.936). Conversely, verbalized introspection collapses on audio input (trap p_{yes} $0.055 \rightarrow 0.556$) on both MiniCPM duplex generations while omni-streaming and frozen-backbone controls stay intact. Three controlled arms (ASR audit, filler audio, audio+text duplication) pin the mechanism to *binding*: the verbal self-check runs over text-token pathways that audio embeddings do not feed, while a probe reads the same information at 0.93+.
4. **End-to-end training creates the shared core (§6).** Text $\leftrightarrow$ audio probe transfer is ~ 0.86 on every end-to-end model but dead (0.52–0.60) on Freeze-Omni’s frozen backbone with identical text weights — so the practical recipe “calibrate on cheap text, deploy on speech” is valid for end-to-end models only.
5. **The gate wins as a system (§7).** The mid-layer probe decides in 20 ms (text) / 45 ms (audio), before the first output token. Against the model’s own `enable_thinking` mode, gated-cloud escalation dominates on both accuracy and latency at every escalation rate, be-

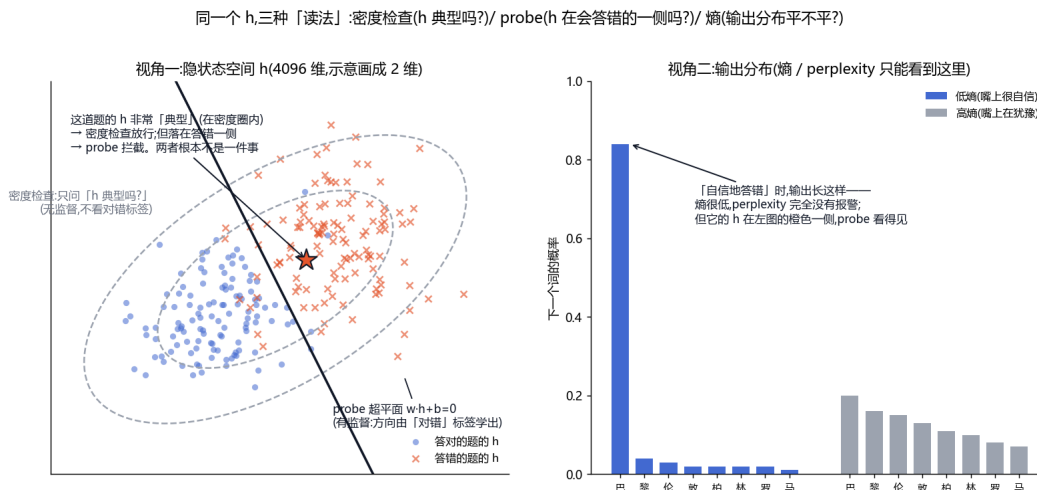


Figure 1: Conceptual contrast between a supervised probe in hidden-state space and what output-distribution scalars (entropy/perplexity) can see. [TODO: regenerate this figure with English labels (current version has Chinese axis/legend text); source: figures/concept_probe_vs_ppl.py]

84 cause the gate flags knowledge/trap failures that thinking cannot fix — external escalation
 85 is *necessary*, not merely better.

86 6. **The loop closes live (§8).** In full streaming duplex sessions, an end-of-turn mid-layer read
 87 out-performs the offline reference (AUC .887 vs. .843) and still decides pre-first-token; the
 88 returned expert answer is spoken by the talker, and the pushback problem largely dissolves
 89 for *gated* escalations because the same low-confidence state that fires the gate predicts
 90 relay compliance (comply .42 → .75 across gate-score quartiles). The live system climbs
 91 monotonically from its honest streaming floor (.400 → .633 at 55% escalation), with the
 92 gap to offline operation itemized into measured audio, streaming, and speech-channel taxes.

93 The paper executes **both steps of the two-step design**. Step 1 (§4–§7) establishes when the talker
 94 should hand off; step 2 (§8) returns the thinker’s result into the live duplex session. Its anticipated
 95 obstacle — the talker pushing back on injected results that conflict with its own reasoning — turns
 96 out to be governed by the gate’s own signal: escalations the gate fires are exactly the ones the talker
 97 is willing to relay, and where confidence is high, authority *framing* works while forcing words into
 98 the talker’s mouth backfires (§8.2).

99 2 Related Work

100 **Do language models know what they know?** Kadavath et al. [2022] introduced $p(\text{True})$ and
 101 $P(\text{IK})$ self-evaluation and showed large models are reasonably calibrated about their own answers;
 102 Lin et al. [2022] trained models to verbalize uncertainty. A parallel line reads correctness from
 103 internal states: linear probes [Alain and Bengio, 2016] on hidden activations separate true from
 104 false statements [Azaria and Mitchell, 2023, Marks and Tegmark, 2024], latent knowledge can be
 105 found without supervision [Burns et al., 2023], and intermediate layers encode more truthfulness
 106 information than the output distribution reveals [Orgad et al., 2025] — consonant with our mid-layer
 107 findings. Output-side uncertainty measures such as semantic entropy [Kuhn et al., 2023, Farquhar
 108 et al., 2024] require sampling multiple generations, which a real-time duplex loop cannot afford.
 109 Our contribution to this literature is not a new signal but a *controlled account of when the standard*
 110 *readouts break*: we show the probe and $p(\text{True})$ readouts fail in complementary, modality- and
 111 architecture-specific ways under duplex fine-tuning, while the underlying information survives.

Routing, cascades, and deferral. Cost-quality routing across LLMs is well studied: FrugalGPT [Chen et al., 2023] cascades models with a learned scorer, HybridLLM [Ding et al., 2024] trains a query router, RouteLLM [Ong et al., 2024] learns routers from preference data, and RouterBench [Hu et al., 2024] standardizes evaluation; LLMRouterBench authors [2026] report that many routers fail to beat simple baselines under a unified protocol and attribute the remaining oracle gap to model-recall failures. These routers decide from *query-side* features before generation, and the large model then answers the user directly. Both assumptions fail in full-duplex voice: the user talks to one voice whose session state (audio context, duplex KV cache) lives in the small model, and our audit (§4) quantifies exactly how far query-side features go — they are the “type shortcut” whose transfer collapses under distribution shift, whereas the model-internal mid-layer signal transfers. Classical learning-to-defer [Madras et al., 2018] and early-exit computation [Schuster et al., 2022] are the same decision inside one model; speculative decoding [Leviathan et al., 2023] inverts our roles (large model verifies small drafts) — our prefetch analysis (§8.1) is its query-level mirror, *speculative escalation*, with the completeness curve as acceptance rate. Reasoning-mode escalation (§7) connects to test-time-compute models [DeepSeek-AI et al., 2025]: we find self-escalation to a thinking mode is dominated by external escalation because the flagged failures are knowledge-bound, not compute-bound.

Full-duplex and omni speech models. End-to-end spoken dialogue models now span duplex designs — dGSLM [Nguyen et al., 2023], Moshi with its Inner Monologue [Défossez et al., 2024], GLM-4-Voice [Zeng et al., 2024] — and streaming omni models such as Qwen2.5-Omni [Xu et al., 2025]; Freeze-Omni [Wang et al., 2024a] attaches speech adapters to a frozen LLM, and Qwen2-Audio [Chu et al., 2024] covers audio-understanding fine-tunes. A related line retrofits full-duplex *behavior* onto a half-duplex generator by adding a dedicated *turn-taking controller* beside or on top of it — perception stays always-on while generation remains half-duplex: FlexDuo [Liao et al., 2025] is an external pluggable module driving a three-state Speak/Listen/Idle FSM at 120 ms intervals; MinMo [Chen et al., 2025] and Freeze-Omni attach lightweight state heads to the LLM’s own hidden states (a single-layer Transformer and a chunk-level classification layer, respectively) to decide when to start or stop speaking. These controllers arbitrate *channel occupancy* — whether the local model should listen or speak next — whereas our gate arbitrates *competence* — whether whoever speaks should be the local model at all. The two decisions are orthogonal and mechanically compatible: two of the three implement their decision as a readout over the same hidden states our mid-layer gate uses, so an escalation head could ride the identical interface. A head-to-head comparison would be ill-posed — they optimize turn-taking F1 on human-human corpora, we optimize adequacy–latency on QA pools — but they serve as two indirect references. First, their arbitration cadences (a decision every 120 ms for FlexDuo, 160–320 ms chunk-level for Freeze-Omni, 250 ms for MinMo) establish what a live duplex loop already tolerates. These budgets are set by *evidence accumulation*, not compute: turn-taking predicts the interlocutor’s behavior, so the controller must wait for audio to disambiguate a backchannel from an interruption (FlexDuo inserts an explicit 500 ms semantic buffer), whereas competence is a property of the model’s own state, available the moment the prompt is encoded. The decision compute itself is near-free for the two head-based designs, exactly as our gate’s readout is (~ 0.03 ms, Table 9); only FlexDuo pays real compute (an extra 7B forward per 120 ms tick). Second, their shared design choice of a *last-layer* readout is one point on our per-layer curve (§7), where the mid-layer point dominates it for competence. Our target, MiniCPM-o 4.5 [Yao et al., 2024, OpenBMB, 2026], is a 9B omni/duplex fine-tune of Qwen3-8B [Yang et al., 2025]. We use this landscape as a *controlled independent variable*: matched pairs separating raw backbones, audio-only fine-tunes, streaming-omni fine-tunes, duplex fine-tunes, and frozen-backbone adapters let us attribute readout damage to duplex-ness specifically, and the shared mid-layer core to end-to-end training specifically.

Evaluation practice. Our query pools come from public benchmarks only: SimpleQA [Wei et al., 2024] for confidently-wrong long-tail traps (cf. PopQA [Mallen et al., 2023]), MMLU-Pro [Wang et al., 2024b], TriviaQA [Joshi et al., 2017], GSM8K [Cobbe et al., 2021] and MATH-500 [Hendrycks et al., 2021, Lightman et al., 2024], and Dolly/Alpaca instruction data [Conover et al.,

Table 1: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

2023, Taori et al., 2023]. Adequacy labels come from an LLM judge [Zheng et al., 2023] validated by a stronger re-judge (agreement 0.962 text / 0.945 audio). The audio arm TTS-renders the frozen pool, following the practice of Spoken-SQuAD [Li et al., 2018] and VoiceBench [Chen et al., 2024]; real-speech validation on SD-QA [Faisal et al., 2021] is planned (§11).

3 Experimental Setup

Target model and matched-pair controls. The system target is **MiniCPM-o 4.5** (9B; omni + full-duplex fine-tune of Qwen3-8B), run in bf16 with greedy decoding, seed 42. To attribute effects to the *kind* of fine-tune rather than the backbone, we assemble matched pairs and controls (Table 1): the raw backbone family member for each fine-tune, a streaming-omni fine-tune (Qwen2.5-Omni-7B), a second-generation duplex model (MiniCPM-o 2.6), an audio-understanding fine-tune (Qwen2-Audio-7B), a frozen-backbone adapter model (Freeze-Omni, whose LLM weights are byte-identical to its raw control), and two additional raw backbones for diversity (Mistral-7B-v0.3 [Jiang et al., 2023], GLM4-9B-chat [Team GLM et al., 2024]); the Qwen2.5/Qwen2 raw controls follow Yang et al. [2024].

Query pool (frozen, public benchmarks only). 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test, seed 42): *trap* = SimpleQA (50; the target model fails 100%), *hard-knowledge* = MMLU-Pro (150), *easy-fact* = TriviaQA (100), *easy-chat* = Dolly-15k (en) + Alpaca-zh (150), *hard-math* = GSM8K tail + MATH-500 (150). Text-input fail rates for MiniCPM-o 4.5: *trap* 1.00 / *knowledge* .48 / *fact* .34 / *chat* .21 / *math* .19 (overall .358). We deliberately use no LLM-generated queries: a pilot GPT-generated trap pool failed its design (90% answered correctly by the 9B target) and creates a generator-equals-judge circularity.

Gate signals (all zero-training). (i) *Decode scalars*: max token entropy / margin over the first k decode steps. (ii) *Linear probe*: logistic regression on a single hidden state, position $\times$ layer swept; trained on calibration features, thresholds from out-of-fold scores. (iii) *Verbalized self-evaluation* [Kadavath et al., 2022]: $p(\text{True})$ -pre reads $P(\text{Yes})$ off the first token of “Would you answer this correctly?” asked *before* answering; $p(\text{True})$ -post asks “Is this answer correct?” about a drafted answer.

Labels and judge. Adequacy labels come from gpt-5.4-mini with structured outputs, blind to the gate. The judge is validated by a gpt-5.5 re-judge: agreement 0.962 (text) / 0.945 (audio); every headline number moves ≤ 0.010 under the stronger judge’s labels. Escalation target for system experiments: gpt-5.5. [TODO: cite/describe judge models properly; OpenAI model documentation entry in refs.bib is a placeholder]

Table 2: Signal AUCs against judged failure labels, MiniCPM-o 4.5, text input. Bracketed interval is a bootstrap 95% CI.

signal	AUC	training needed
$p(\text{True})$ -post (post-draft)	0.899	none
probe on h_{prompt} (final layer)	0.822 [0.777, 0.863]	LR on calib
$p(\text{True})$ -pre (pre-answer)	0.807	none
max entropy@4 (best decode scalar)	0.696	none

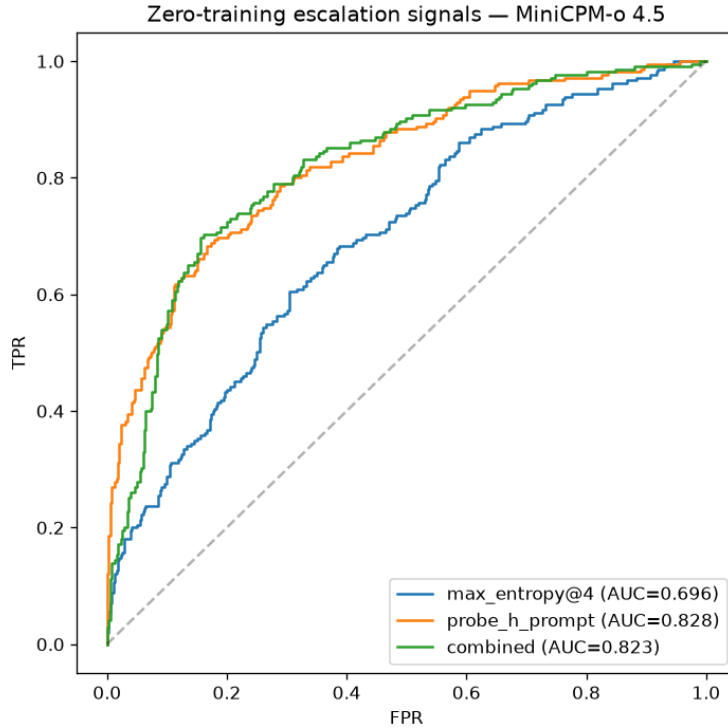


Figure 2: ROC of the final-layer probe vs. decode scalars (calibration split). **[TODO: regenerate to include the $p(\text{True})$ curves on the same axes]**

196 **Audio arm.** The frozen pool is TTS-rendered (OpenAI tts-1, single voice), so query *content*
 197 is unchanged and only modality varies, following Spoken-SQuAD / VoiceBench practice [Li et al.,
 198 2018, Chen et al., 2024]. Real-speech validation on SD-QA [Faisal et al., 2021] is open (§11).

199 **Infrastructure.** All experiments run on Modal H100 instances; torch 2.8.0, transformers 4.51.0
 200 (4.57.6 for Qwen2.5-Omni; 4.45.2 for Freeze-Omni). Total compute + API spend for Phases 0–7a:
 201 ≈\$290.

202 4 Which Signal Knows the Model Will Fail? (Text Input)

203 4.1 Headline comparison

204 Table 2 compares the candidate signals on MiniCPM-o 4.5 (calibration split; probe numbers are
 205 5-fold out-of-fold). Figure 2 shows the ROC curves.

Table 3: LOPO transfer (AUC on the held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched. The o4.5 column is the Qwen3-family pair for reference.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6 (duplex)	[o4.5 vs Qwen3]
hard-math	0.809	0.745	0.538 $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	0.526 $\approx$ chance	0.61

4.2 The probe’s number is a type shortcut

The probe’s 0.822 does not mean what it appears to mean. Three audit results (all calibration-only): (i) the same hidden state classifies the *source pool* of a query at 95.8%; (ii) a pool-identity oracle alone — predict each pool’s base fail rate — already reaches AUC 0.715; (iii) appending true pool labels to the probe adds nothing (0.822 $\rightarrow$ 0.821), i.e. the probe has already absorbed the type information. The decisive test is leave-one-pool-out (LOPO) transfer: held-out math *inverts* to 0.372 — the probe scores terse MATH-500 items as “safer” than verbose GSM8K items, a style-familiarity artifact — and the mean escalation score on the 100%-fail trap pool is 0.232. Decode entropy is $\approx$ chance on two of three raw backbones tested. In-distribution AUC therefore overstates what a query-side signal knows; we treat LOPO as the primary metric for signal quality throughout.

4.3 $p(\text{True})$: ask before the model answers

Verbalized self-evaluation needs no training and transfers differently. On the 100%-fail SimpleQA trap pool, pre-answer $p(\text{True})$ says “No, I will not get this right” with mean escalation score 0.945 — on the same items the probe scores 0.232. The timing is essential: asked *after* the model has produced a confident-wrong answer, self-assessment collapses to 0.360 (the model endorses its own answer). Post-draft $p(\text{True})$ (“Is this answer correct?”) is the strongest overall signal (0.899) but requires paying for a full draft first. Across backbones, $p(\text{True})$ -post beats the trained probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, 0.685 vs. probe 0.798) shows the pattern depends on the backbone’s self-evaluation calibration quality rather than being universal.

5 Duplex Fine-Tuning Overwrites Readouts, Not Knowledge

5.1 Matched pairs: damage tracks duplex-ness, not audio

Is the LOPO transfer failure of §4.2 a MiniCPM quirk? Table 3 shows LOPO transfer of the standard (final-layer, last-token) probe across one backbone family, with label coverage matched by subsampling the raw control to the duplex model’s per-pool fail rates (`lopo_matched`): the raw backbone matched to MiniCPM-o 2.6’s exact fail rates keeps math at 0.825 while the duplex model itself sits at chance — so label coverage is ruled out and the *representation* changed. The degradation is ordered: raw $>$ omni-streaming $>$ duplex. Audio-only fine-tuning is separately controlled (Qwen2-Audio), though that pair is confounded by capability collapse and is reported as a documented negative. Raw-backbone baselines vary across families, so the defensible statistic is always the within-pair difference.

5.2 Layer $\times$ position sweep: the cliff

We capture every decoder layer at prefill, at both the last-token and mean-pooled positions (final-layer values reproduce the generation-time hooks exactly, 5-for-5). Figure 3 and Table 4 show the central result: **mid-network, the duplex model carries near-raw transferable difficulty information; the damage is confined to the last few layers at the last-token position.**

Three structural facts localize the mechanism. First, the within-pair deficit at the best mid-layer is only -0.03 , vs. -0.59 at the final layer. Second, mean-pooled probes survive to the final layer on duplex models — the damage is (late-layer $\times$ last-token)-specific, exactly the position a streaming/duplex head must re-purpose to encode turn-control state. Third, the two fine-tune

Table 4: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	0.931 (L22/36)	0.366	cliff L31→L34, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse mild depression
MiniCPM-o 2.6 (duplex)	0.822 (L21/28)	0.540	cliff L21→L27

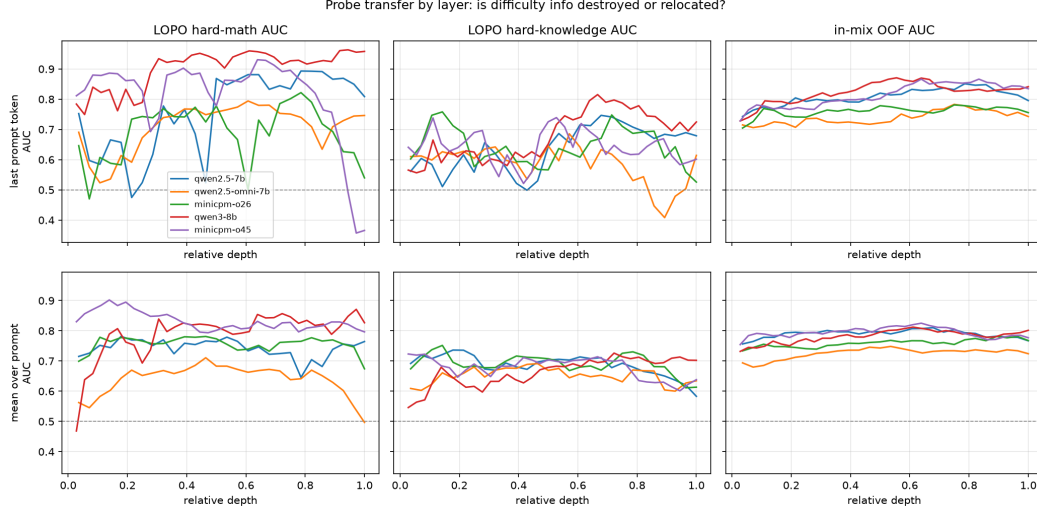


Figure 3: LOPO transfer by layer and position. Duplex models (MiniCPM-o 4.5/2.6) show a late-layer cliff at the last-token position only; mean-pooled probes survive to the final layer; raw backbones show no cliff; the omni-streaming control shows mild diffuse depression instead.

245 styles fail differently: duplex damage is severe but *local*; omni-streaming damage is mild but *diffuse*.
246 We conclude the duplex fine-tune **overwrites the readout** rather than destroying the model’s self-
247 knowledge — which reverses the natural interpretation of §4.2: the signal exists and is strong; the
248 standard readout simply cannot see it on duplex models.

249 5.3 Does the mid-layer convert into a better gate?

250 On the frozen test split, accuracy-vs-escalation-rate areas over the random baseline order as: final-
251 layer probe $+0.054 < p(\text{True})\text{-pre} +0.059 < \textbf{mid-layer L22 probe} +0.064 < p(\text{True})\text{-post}$
252 $+0.068$ (which requires a full draft). The optimum is not knife-edge (L20–L30 all $\geq +0.056$; mean-
253 pool uniformly worse), and the in-mix area *under-sells* the mid-layer probe: its real advantage is
254 LOPO robustness (math 0.93 vs. 0.37), which a same-mix test cannot show (Figure 4). Caveat:
255 quantile-threshold transfer of probe scores remains weak and needs deployment-grade calibration
256 (§11); AUC/area claims are unaffected.

257 5.4 A second duplex family: the pre-registered test executed

258 The mid-band claim was pre-registered as a prediction: when another open-weight full-duplex fam-
259 ily became available, its usable read should also sit mid-network, not at the standard (final-layer,
260 last-token) readout. NVIDIA NemotronLabs-VoiceChat-11B — a hybrid Mamba/attention back-
261 bone (Nemotron Nano v2 9B: 56 blocks, 27 Mamba-2 / 4 attention), i.e. architecturally as far from
262 MiniCPM’s dense transformer as the open-weight space currently offers — provides the test, and
263 it passes on first execution. With the identical recipe (same wavs, same judge, 600-query cali-
264 bration only), the layer sweep peaks mid-band at L30–34 (eot-last AUC .714 vs. .693/.682 at the

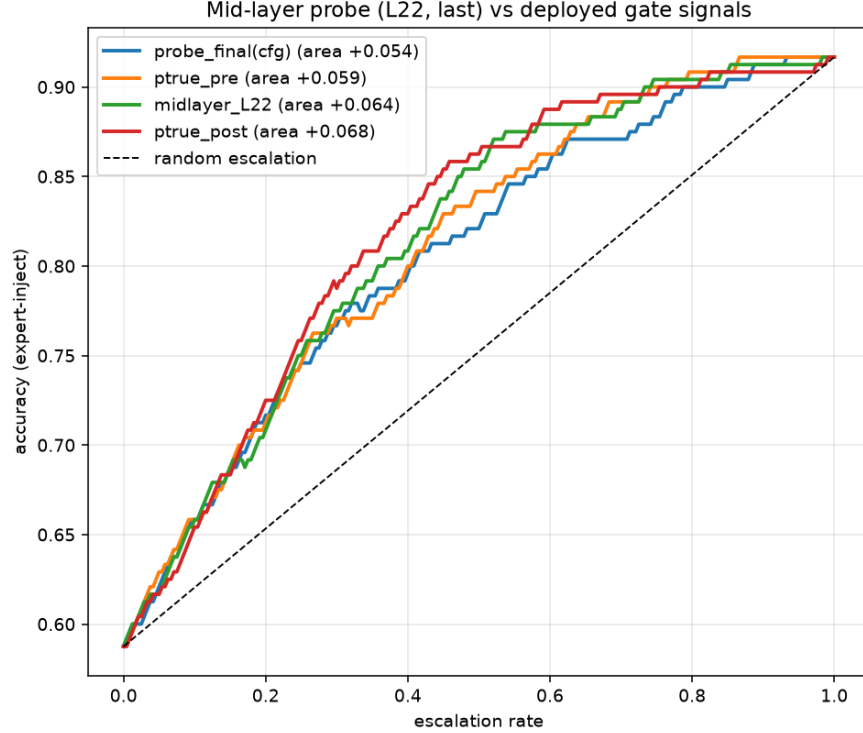


Figure 4: Accuracy vs. escalation rate on the frozen test split for the mid-layer (L22) gate against final-layer probe and $p(\text{True})$ variants.

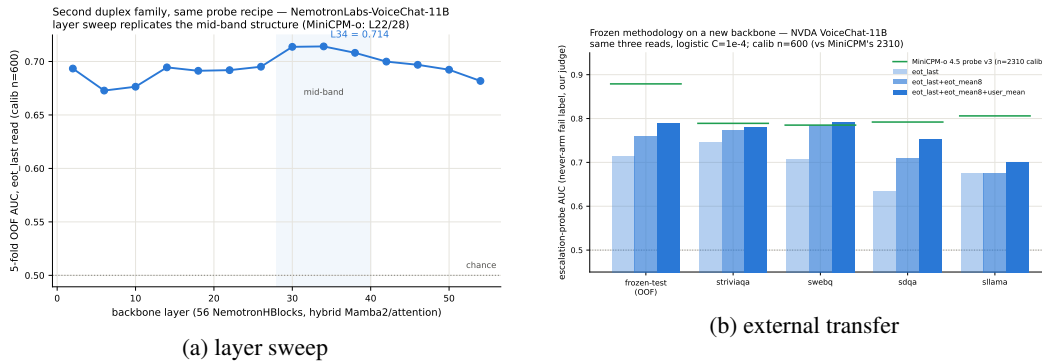


Figure 5: The pre-registered second-family test on NemotronLabs-VoiceChat-11B (hybrid Mamba). (a) The usable read again peaks mid-network (L30–34 of 56). (b) The frozen-recipe probe transfers to the external pools within the MiniCPM v3 AUC band.

265 endpoints); stacking the same three zero-latency positions lifts OOF .714 $\rightarrow$.761 $\rightarrow$.790; and
 266 the frozen-methodology probe transfers to the external pools at AUC .781/.793/.754/.701 (strivi-
 267 aqa/swebq/sdqa/sllama) — the MiniCPM v3 band reached with a quarter of the calibration data
 268 (Figure 5). Two boundary notes: the Nemotron backbone is substantially weaker at knowledge
 269 retrieval (striviaqa local accuracy .32 vs. MiniCPM’s .62, same judge), and it is English-only —
 270 Chinese audio yields fluent unrelated English, so the cross-lingual pool has no analog here (zero la-
 271 bel variance; skipped). The live 4-arm port and the trained-checkpoint ablation remain future work;
 272 the replicated claim is the structural one: *mid-network readability of the self-assessment signal is a*
 273 *property of duplex models as a class, not of one checkpoint.*

Table 5: Pre-answer introspection under audio input. Trap p_{yes} is the mean stated probability of answering correctly on a pool the model fails 100% (lower is better-calibrated).

model	FT type	trap p_{yes} : text $\rightarrow$ audio	audio $p(\text{True})$ -pre AUC
MiniCPM-o 4.5	duplex	.055 $\rightarrow$.556 (collapse)	.786 (trap dead)
MiniCPM-o 2.6	duplex	.196 $\rightarrow$.345 (compresses to $\sim$.5)	.491 $\approx$ chance
Qwen2.5-Omni	omni-streaming	.279 $\rightarrow$.213 (intact)	.727
Freeze-Omni	frozen + adapter	.165 $\rightarrow$.112 (intact)	(.612; capability caveat)

6 Speech Input: Each Readout Has One Blind Modality

6.1 The late-layer cliff is text-input-specific

Running the identical layer sweep on the TTS-rendered pool (audio input, content unchanged): audio $\rightarrow$ audio LOPO math reaches 0.93–0.96 at mid-layers *and the final layer does not invert* (L35 = 0.936, vs. 0.366 on text). A speak-mode template control (use_tts_template=True on text input) leaves the cliff unchanged (L35 = 0.362) — the operative variable is the modality of the *input context* (text tokens vs. audio embeddings), not the output mode. The duplex fine-tune re-purposed late-layer last-position processing of text-token contexts specifically.

6.2 Cross-modal transfer works — for end-to-end models only

Probes trained on text hidden states and scored on audio hidden states (and vice versa) reveal where the representation becomes modality-shared: early layers are modality-specific (0.54–0.60), and from $\sim$ 50% depth text $\rightarrow$ audio transfer plateaus at 0.82–0.87 (0.855 at the deployed L22) on MiniCPM-o 4.5. This replicates on MiniCPM-o 2.6 (onset $\sim$ 46% depth) and cross-family on Qwen2.5-Omni (onset $\sim$ 25%). The decisive control is **Freeze-Omni**: identical frozen Qwen2-7B weights with audio attached via an adapter — and cross-modal transfer is $\approx$ dead (text $\rightarrow$ audio 0.52–0.60, audio $\rightarrow$ text 0.34–0.54). **The modality-shared mid-layer core is created by training the backbone on the modality**; an adapter aligns well enough to converse, but its audio-context hidden states live off the text manifold. Practically: the cheap recipe “calibrate the gate on text data, deploy on speech” is validated for end-to-end models and invalid for adapter-attached ones.

6.3 Verbalized introspection collapses on audio — and why

The mirror-image failure: pre-answer $p(\text{True})$, robust on text (§4.3), collapses under audio input on both MiniCPM duplex generations, while both non-duplex controls stay intact (Table 5).

Three controlled arms pin the mechanism on MiniCPM-o 4.5:

1. *Not perception*. An ASR audit shows the model hears the trap questions fine (WER .074); $p(\text{True})$ computed on the model’s *own transcript* snaps back to .074 $\approx$ text; the well-heard subset still collapses (.582); corr(WER, p_{yes}) $\approx$ 0.
2. *Not a context prior or persona*. Irrelevant filler audio alongside the text question does *not* inflate p_{yes} (.001 — it depresses it everywhere).
3. *Binding*. Duplicating the question as text tokens alongside the audio fully restores introspection (.034). The verbal instance-check runs over **text-token pathways; audio embeddings do not feed it** — while a linear probe reads the same instance information off the same forward pass at 0.93+.

A log-odds decomposition agrees: the audio shift is graded (chat +0.19 $\rightarrow$ trap +4.37), i.e. on audio input verbal self-assessment regresses to type-level priors. The failure also implies its own cheap fix, confirmed twice: re-present the query as text (own transcript or duplication) before asking — “repeat-then-judge.”

Table 6: Mechanism synthesis across training regimes.

	shared mid-layer core	faithful readouts
raw text backbone	(text only)	✓
omni-streaming e2e (Qwen2.5-Omni)	✓built	✓intact
duplex e2e (MiniCPM-o × 2)	✓built	× overwritten (probe: late-layer/text; verbal: audio)
frozen backbone + adapter (Freeze-Omni)	× absent	✓ verbal survives (capability craters instead)

6.4 Real-speech validation: not TTS artifacts

Every audio number above uses TTS speech (one voice). We rerun the matched-pair design on 200 *real human recordings* (VoiceBench SD-QA, USA split; naturally-occurring factoid questions with references), each question presented both as typed text and as the human recording. All three audio-side findings replicate: the modality tax (fail rate .400 → .450); audio overconfidence (paired p_{yes} shift +.089; on the failure subset .415 → .581); and the layer structure — early layers dead ($\leq L11$, .42–.55 across all transfer arms), a sharp rise after ~50% depth, audio late layers usable through L35 (.74–.77), and the deployment recipe (text-calibrated probe read on real-speech audio) holding a .76–.80 band at L22–L26 (peak .800 at L25). Magnitudes sit below the matched-TTS .86 as expected: SD-QA content is also out-of-pool, making this transfer across *both* content and modality — a strictly harder test. The text-side math cliff is untestable here (SD-QA contains no math); it never involved synthetic audio and needed no such protection. Residual limitation: scripted read speech in one dialect (ten further dialect splits remain for future robustness work).

6.5 Synthesis: four quadrants

End-to-end multimodal training *builds* the shared semantic core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (Table 6). There is an elegant symmetry in the two failures: the probe’s readout is text-fragile and $p(\text{True})$ ’s readout is audio-fragile — in both cases the knowledge survives and a *readout* breaks, and in both cases the breakage is duplex-fine-tune-specific. For gate design the consequence is concrete: **the mid-layer last-token probe is the one signal that survives every quadrant that can converse.**

7 The Gate as a System

7.1 Accuracy–cost tradeoff

On the frozen test split ($n = 240$): small-only accuracy **0.588**; big-only (gpt-5.5) **0.917**; big relayed through the small model’s paraphrase 0.879 (relay tax 1–4 points). Every gate beats random escalation at every operating point (Figure 7); the balanced tier at a 33% escalation budget reaches **0.779–0.787**, recovering ~58% of the small→big gap at one third of the big-only cost (\$1.12 per 100 queries big-only).

Figure 7 also draws the baseline the type-shortcut audit (§4.2) demands: a *pool-oracle* router that escalates by each pool’s calibration-split base fail rate (true type labels, no instance-level information, no test leakage). It captures +0.042 of the gate’s +0.054 area over random — the system-level counterpart of the AUC decomposition (0.715 of 0.821), and slightly harsher: in-distribution, the instance-level residual the internal signal adds on top of type routing is +0.012 area (~2 points at the balanced budget) — not significant at $n=240$ (paired bootstrap over test queries, $B=10^4$: 95% CI $[-0.003, +0.027]$, two-sided $p=.13$); in distribution, the gate is statistically on par with the pool oracle. The gate’s case over type routing therefore rests not on an in-distribution margin but on transfer: the type shortcut is exactly what collapses under LOPO (§4.2), while the mid-layer signal survives it.

A *trained* query-feature router closes this argument with data. We fit a RouteLLM-style external router [Ong et al., 2024] under identical data conditions — TF-IDF (word 1–2 gram + character

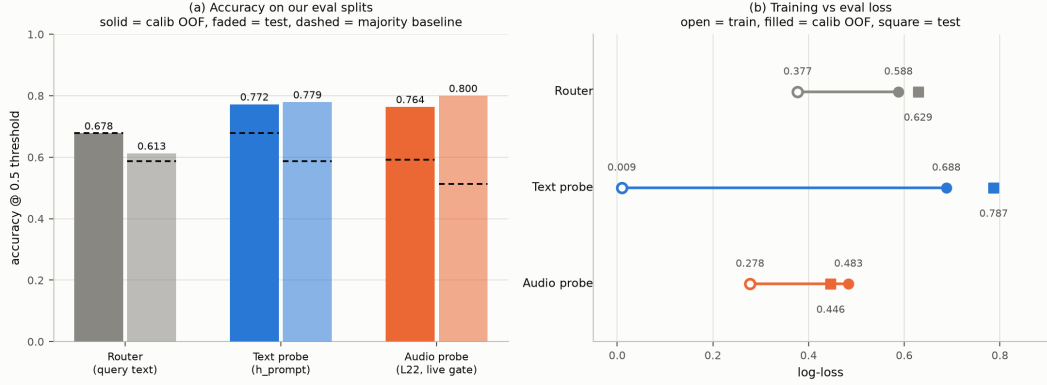


Figure 6: The same training receipt for the external router and the two internal probes (identical labels, identical logistic-regression machinery). (a) Accuracy at the 0.5 threshold on the calibration (out-of-fold) and frozen test splits, against each split’s majority-class baseline (dashed): the router never clears majority; both probes clear it by 9–29 points. (b) Train vs. eval log-loss: the audio probe generalizes cleanly (.278 train / .446 test, the best-calibrated signal); the text probe memorizes in-sample (4096 dims $\gg$ 360 queries) yet its held-out accuracy holds — all reported numbers are out-of-fold or test.

349 3–5 gram) $\rightarrow$ logistic regression on the calibration split’s text labels, seeing only the query sur-
 350 face. In-mix it lands *below* even the pool oracle (OOF AUC .669 vs. .715; tradeoff area +.040 vs.
 351 the probe’s +.054, mid-layer +.064, $p(\text{True})$ -post +.068 on the same test set and frozen expert
 352 answers); a 24-configuration sweep over feature space and regularization tops out at .689, so the
 353 shipped configuration is the query surface’s ceiling, not a tuning artifact. Under LOPO it collapses
 354 entirely (held-out-pool AUC .38–.57 vs. the mid-layer probe’s .93 on math) and its mean score on
 355 the 100%-fail trap pool is .230 — it *misses* the pool a router most needs to catch. The training receipt
 356 makes the point starkly: at the argmax threshold the router’s out-of-fold accuracy on the calibration
 357 split, .678, *exactly matches the majority-class rate* (test .613 vs. majority .588) — 360 queries of
 358 surface text buy a weak ranking signal (train log-loss .38 vs. out-of-fold .59) and no usable classi-
 359 fier. The internal probes pass the same accounting: at the same threshold, same labels, and same
 360 logistic-regression machinery, the text probe’s out-of-fold accuracy is .772 (majority .678) and .779
 361 on test (majority .588), and the live audio mid-layer probe reaches .764/.800 against .592/.512 —
 362 real classifiers, not merely rankers; the difference is purely the input representation (Figure 6).

363 Nor is the baseline data-starved. The identical recipe trained on RouterBench [Hu et al., 2024] —
 364 36,497 public prompts, routing mixtral-8x7b $\rightarrow$ gpt-4 (escalation base rate .432) — reaches out-
 365 of-fold AUC .710 in-domain (accuracy .660 vs. majority .568; deferral area over random +.033):
 366 a hundred times the data buys .04 of AUC, landing at our pool-oracle’s level and still short of
 367 the probe’s .828. Its leave-one-benchmark-out transfer sits at chance on the format-disjoint bench-
 368 marks (HellaSwag .502, GSM8K .509, Winogrande .498), reproducing our LOPO collapse on pub-
 369 lic data, and our calibration-trained router transfers to RouterBench *below* chance (AUC .440). The
 370 transfer failure is symmetric: RouteLLM’s *released* preference-trained routers (BERT and matrix-
 371 factorization, $\sim$ 100k GPT-4-vs-mixtral preference pairs), scored zero-shot on our pool by their own
 372 inference code, reach test AUC only .523/.533 — below even the 360-sample same-data router —
 373 and both rank the 100%-fail trap pool *below* the ordinary hard pools. The audio modality narrows
 374 but does not close the gap: against the live gate’s audio labels the same recipe reaches OOF AUC
 375 .743 / test .814 (gold text; the talker’s own ASR transcript costs it only $-.005$ –.009) — stronger than
 376 on text labels, because the audio channel makes hard pools fail more uniformly and the type shortcut
 377 buys more — yet the audio mid-layer probe still leads every readout (.843/.879 AUC, accuracy .800
 378 vs. .696), and the router still requires the full utterance plus transcription, so it cannot fire mid-stream.
 379 Query-feature routing is the type shortcut and nothing else; the standardized LLMRouterBench pro-
 380 tocol [LLMRouterBench authors, 2026] remains future work, but the information-source question

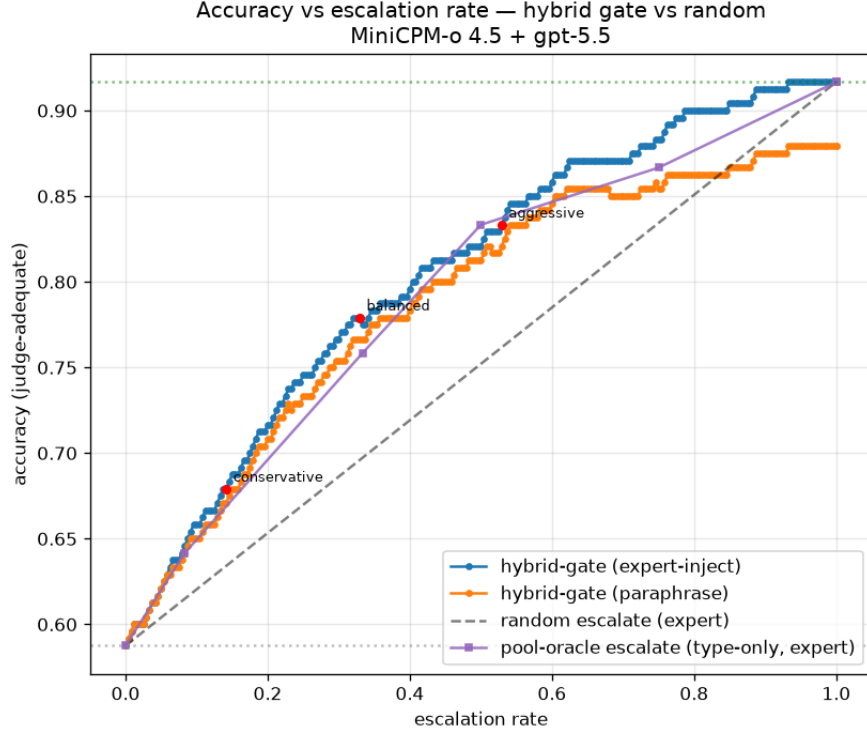


Figure 7: Accuracy vs. escalation rate, frozen test split: gated escalation vs. the random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints. The oracle captures most of the gate’s in-distribution lift (+0.042 of +0.054 area); the gate’s advantage over type routing is transfer, not this margin (§4.2). See also Figure 8 for the $p(\text{True})$ variants.

Table 7: Policy table, frozen test split ($n = 240$). Latency from per-pool measured medians; think and cloud latencies per-query.

policy	acc	lat. mean (s)	P50	P95
fast-only	.588	3.0	3.5	4.2
all-THINK (o4.5’s own)	.637	22.4	17.2	60.1
gated-think @.33	.613	12.1	3.6	47.6
gated-cloud @.15	.688	5.3	3.6	10.1
gated-cloud @.33	.787	6.5	3.6	20.8
gated-cloud @.50	.858	7.0	3.6	24.4

is settled under identical data, the ceiling replicates at $100\times$ the data on a public benchmark, and routing knowledge fails to port across talkers in either direction.

7.2 Against the model’s own thinking mode

MiniCPM-o 4.5 ships a built-in `enable_thinking` reasoning mode — self-escalation in place. Table 7 compares deployment policies with measured latencies. **Gated-cloud escalation dominates all-THINK on both axes at every escalation rate.** The reason is mechanistic, not incidental: the gate predominantly flags knowledge/trap failures, which thinking cannot fix — thinking helps math (fail .187 $\rightarrow$.127) but *hurts* chat (.207 $\rightarrow$.233) — so external escalation is necessary, not merely better.

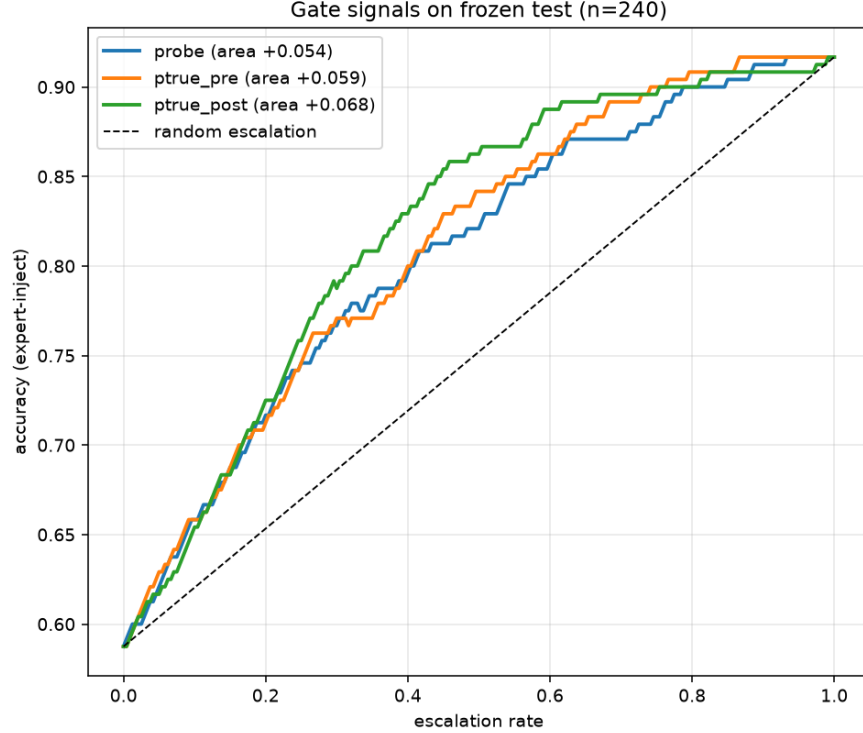


Figure 8: Same tradeoff for the $p(\text{True})$ signals (pre-answer and post-draft).

390 7.3 The fork: deciding before the first token

391 The mid-layer probe is one 4096-d dot product on a hidden state the forward pass already computes:
 392 forking a decision branch at layer k costs microseconds and never stalls the main forward. The
 393 design question is at which depth the signal is sufficient. Measured on H100 (CUDA-synchronized,
 394 truncated-forward): the L22 decision lands in **20 ms (text) / 45 ms (audio)**, vs. time-to-first-token of
 395 36 / 68 ms — the decision exists *before decoding starts* and sits far inside the 200–300 ms turn-taking
 396 budget [Stivers et al., 2009]. Per-layer curves (Figure 9) show L22’s hidden state is available at 57%
 397 of prefill wall-time (61% audio) while L22 is simultaneously the in-mix quality peak (OOF .866 >
 398 final layer .835): deciding early costs nothing, and the cloud call can launch while the remaining
 399 ~40% of prefill plus the whole decode still runs. Forking much earlier is not free: pre-50%-depth
 400 layers are both weaker (L10 $\approx .794$) and modality-specific (§6.2); the fork belongs at ~55–60%
 401 depth. For comparison, $p(\text{True})$ -pre costs 39 / 67 ms (an extra short prefill) and $p(\text{True})$ -post costs
 402 a full draft (1.9 / 3.5 s).

403 7.4 Does the cloud round-trip hide inside the talker’s floor time?

404 If the gate fires at prefill and the cloud call launches immediately, the expert result is ready before
 405 the local answer finishes speaking for 40–58% of the whole test mix (audio input buys more time
 406 than text). But *gate-escalated* queries skew toward short local answers (trap and knowledge pools;
 407 trap overlap ≈ 0), so pure overlap covers only ~20–31% of escalations at a 33% budget (Figure 10).
 408 The residual wait after the local answer ends is P50 2–3 s — one stall sentence — but P90 ~27 s,
 409 dominated by the expert’s own reasoning latency. This argues for a fast-expert tier or streamed
 410 partial results in step 2, and it quantifies the stall-phrase budget an injection protocol must cover.

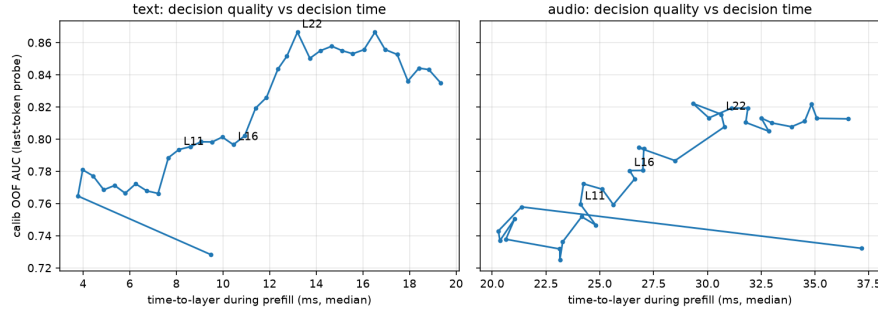


Figure 9: Fork-depth Pareto: per-layer truncated-forward wall-time vs. gate quality. The deployed L22 sits at the quality peak while its hidden state is available at $\sim 57\text{--}61\%$ of prefill wall-time.

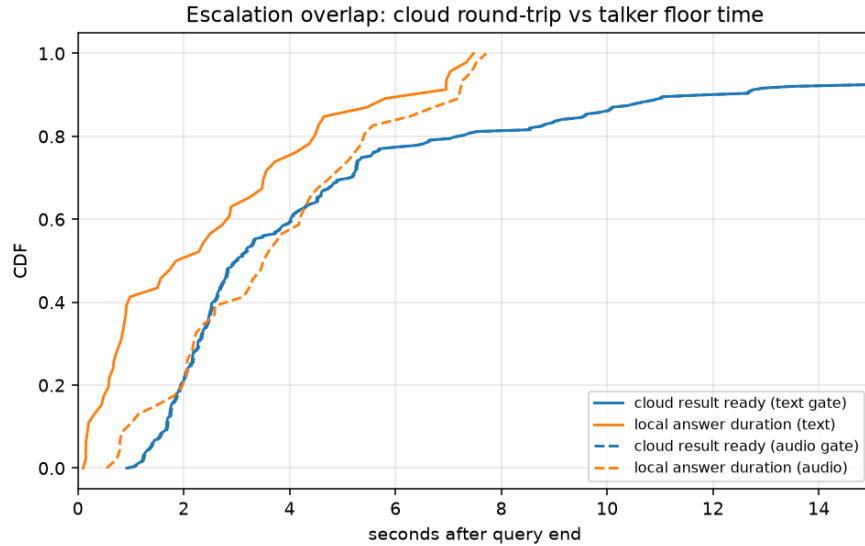


Figure 10: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends (+slack), whole mix and gate-escalated subset, text and audio.

411 8 The Gate Goes Live: Streaming Deployment and Result Injection

412 Everything so far measures signals in replay. This section closes the loop — the original third
 413 research question: once the gate has decided the answer belongs to the big model, how does the big
 414 model’s result get back into the small model so the *talker* speaks it? We run the full duplex session
 415 live: audio streamed in 1 s chunks, the mid-layer gate read during streaming, the expert called mid-
 416 conversation, its result injected into the same session. Three questions structure the section: where
 417 the streaming gate should read (§8.1), whether the talker will actually say what the expert answered
 418 (§8.2), and what the tradeoff curve looks like when every oracle shortcut is removed (§8.3).

419 8.1 The streaming loop: the end-of-turn read decides, prefetch dies

420 The offline gate reads h_{prompt} — the last prompt token’s mid-layer state. Streaming has no such
 421 token until the turn ends, and per-chunk probe scores are strictly weaker than the offline reference
 422 (best chunk statistic AUC .764 vs. offline .843), with a structural failure on trap: short utterances
 423 back-load the critical entity into the zero-padded tail chunk, and the mid-stream gate fired on only
 424 1/5 of trap (live) where the offline read caught nearly all. The fix is the **end-of-turn read**: after the
 425 last audio chunk, prefill an assistant turn containing a single space and read L22 at the assistant-start

position — the streaming analogue of h_{prompt} . This read reaches **AUC .887**, *above* the offline full-prefill reference (.843), restores trap fire@balanced from .20 to **.90**, costs 22 ms (still before the first output token), and becomes the *only* decision point — escalation-rate thresholds are quantiles of this single score, so the chunk-vs-turn joint-calibration problem disappears.

The remaining role for mid-stream firing would be *speculative* expert prefetch — start the cloud call early, veto at end of turn. We kill this with a measurement before building it. Framing the design as *speculative escalation* — speculative decoding’s draft–verify skeleton [Leviathan et al., 2023] applied to the escalation decision itself, at sentence granularity — its acceptance rate is the probability that an expert answer to the *partial* transcript answers the full question — a wait- k style tradeoff [Ma et al., 2019]. Judged on calibration texts truncated at 25/50/75% of the words, acceptance is **.07/.19/.51**; at the chunk gate’s typical fire point (27–40% of the audio) it is $\sim .2$ –.4, i.e. 6–8 of every 10 speculative calls would be wasted. Trap’s acceptance floor (.00/.10/.27) is the same back-loaded-core property that broke the mid-stream probe — one underlying property, measured independently in the hidden states and in the semantics. Prefetch is dropped; the expert starts at end of turn.

Two more deployment-honesty decisions close the loop’s design. First, the expert query is the talker’s *own* transcription of the audio (P50 2.2 s, overlapped with the stall phrase), not the dataset’s gold text. Second, expert reasoning effort is **fixed at low**: on the gate-escalated population, effort buys *zero* accuracy (gold-text queries: .85 medium vs. .82 low overall; math 1.00 at both) while doubling latency (P50 5.8→9.1 s, P95 23.9→78.8 s) — escalated failures are knowledge-bound at both scales, mirroring the talker-side thinking result (§7.2). A planned deadline-aware effort policy collapsed, by measurement, to the simplest possible one. The stall phrase itself must be injected as *assistant-role prefill*: the official `teacher_forcing_text` hook continues generating freely mid-stream, so canned filler semantics survive only if the phrase enters as already-spoken assistant speech.

8.2 Will the talker say it? The pushback problem, quantified

The streaming smoke test surfaced the substantive step-2 obstacle: the talker *pushes back* on injected expert results that conflict with its own reasoning. Before any live sweep, we quantified this channel on 172 live audio sessions: conflicting “expert answers” fabricated by within-pool derangement of reference answers (deterministic, no LLM generation), four framings from neutral (F0) through the deployed authority template (F1), strong-override (F2), and assistant-seeding (F3, teacher-forcing “the answer is: X” into the talker’s own mouth), judged into {comply, pushback, lip-service, other}. Five results:

1. **The deployed channel is clean.** On the actual live workload — model wrong, correct answer injected — comply = 1.00 ($n=24$). The live curve of §8.3 is not contaminated by relay refusal.
2. **The same signal that triggers escalation predicts compliance.** Across gate-score quartiles, comply rises $.42 \rightarrow .67 \rightarrow .75 \rightarrow .75$; resistance (pushback + lip-service) is .25 in the most-confident quartile vs. .08 elsewhere. A model that knows it doesn’t know is willing to listen — the gate and the relay are two uses of one signal. In the high-score half, hard-knowledge (the dominant escalation type) complies 1.00 ($n=13$).
3. **Authority framing works; forcing words into the talker’s mouth backfires.** Worst case (confident-correct model $\times$ conflicting injection): F0 comply .00, F1 .53, F2 .79 — but F3 *seeding* yields comply .25 with lip-service .62: the model speaks the seeded answer, then walks it back in its own continuation. Forced speech is treated as revisable; framing, not ventriloquism, is the mechanism. The deployed loop keeps F1 (F2 is the shelf option: it only helps a rare corner and plausibly raises the swallow rate below).
4. **Math is silently re-derived.** “Other” concentrates in hard-math: the model neither relays nor disputes — it re-derives and states its own number (comply .57 / silent .26 vs. .66 / .20 non-math). A bare numeric answer carries no authority against the model’s own

Table 8: Live streaming sweep, probe v3, paired bootstrap 95% CIs ($B=10^4$). Top block: speakable subset ($n=218$, headline); bottom: full frozen test pool ($n=240$). Heard = honest end-to-end accuracy; gold-inject = the channel-controlled counterfactual (escalated rows re-scored with the frozen gold-text expert answers); channel cost = their paired difference. * = CI excludes 0; † = CI lower bound at zero.

tier	esc	heard-acc	Δ vs. floor	gold-inject	channel cost
never (floor)	0%	.422 [.36,.49]	—	.422	—
conservative	14%	.528 [.46,.59]	+106* [+05,+17]	.550 [.49,.62]	+023† [.00,+05]
balanced	31%	.592 [.53,.66]	+170* [+11,+24]	.661 [.60,.73]	+069* [+03,+11]
aggressive	57%	.633 [.57,.70]	+211* [+14,+28]	.761 [.70,.82]	+128* [+08,+18]
always (synth)	100%	—	—	.922 [.89,.95]	—
never (full)	0%	.383 [.32,.45]	—	.383	—
conservative	16%	.483 [.42,.55]	+100* [+05,+15]	.525 [.46,.59]	+042* [+02,+07]
balanced	35%	.554 [.49,.62]	+171* [+11,+23]	.654 [.59,.71]	+100* [+06,+14]
aggressive	61%	.596 [.53,.66]	+212* [+15,+28]	.771 [.72,.82]	+175* [+13,+23]
always (synth)	100%	—	—	.917 [.88,.95]	—

476 computation chain; harmless when the expert is right, but verbatim-relay claims are off the
477 table for math.

478 5. **The swallow rate sets the ceiling.** Model-wrong $\times$ wrong-injection: comply .64, push-
479 back .00. An unconfident model has no internal resistance to a wrong expert — expert
480 quality is the accuracy ceiling of the whole cascade, now quantified.

481 8.3 The live curve, with the deployment price itemized

482 With the channel verified clean, we ran the full live sweep: 240 frozen test queries $\times$ four tiers (never
483 / conservative / balanced / aggressive), each a complete duplex session with the deployed v3 probe
484 (the multi-position L22 refit of §9), scored on *heard accuracy* — the judge sees exactly what a user
485 would have heard (the relay for escalated queries, the local streaming answer otherwise). Labels
486 are frozen in `frozen_v3_traces.parquet`; all statistics are paired bootstrap over queries on those
487 labels (Table 8, Figure 11).

488 Headline numbers are reported on the **speakable subset** ($n=218$): a pre-registered input-side fair-
489 ness filter — decidable from query text and wav bytes alone, never from outcomes — removes 21
490 formula-LaTeX ids and one digitally-silent wav from the 240. Formula content read aloud by a TTS
491 is destroyed before the system under test ever hears it (no one speaks LaTeX at a voice assistant),
492 and the exclusion is asymmetric in exactly the way the artifact predicts: the gold-inject ceiling barely
493 moves (.917 $\rightarrow$.922) while the aggressive channel gap drops .175 $\rightarrow$.128 — LaTeX rows destroy
494 the speech channel specifically, they are not intrinsically harder for the expert. Rare entities and hard
495 names are deliberately *not* filtered: mishearing them is the phenomenon. Full-pool rows appear in
496 Table 8 for comparison.

497 The curve rises monotonically and every tier’s gain over the live floor is significant:
498 +.106/+.170/+.211 on the speakable subset, +.21 at 61% escalation on the full pool. Selectivity, how-
499 ever, must be read in the channel-controlled view. In the heard view the higher tiers sit at or below a
500 random-escalation reference drawn to the gold ceiling (aggressive .633 vs. .709): the speech-channel
501 cost (+.128 at that tier) eats the routing margin, so on this mixed pool escalating *at all* helps, but
502 *which* queries the gate picks is masked by the uplink. Remove exactly that channel — the gold-inject
503 view — and the gate clears the same random reference at every arm (+.059/+.083/+.052). The rout-
504 ing signal survives deployment; the binding constraint at high escalation rates is the transcription
505 uplink, not the gate (the external pools of §9, where the channel is milder, show the heard-view ad-
506 vantage directly). One deployment correction fell out of this analysis: re-deriving the tier thresholds
507 as label-free quantiles of the deployment-mix scores (`gate_v3_thresholds_corrected.json`)
508 removes 11.3% of the expert calls at equal accuracy on this pool — the drifted aggressive threshold

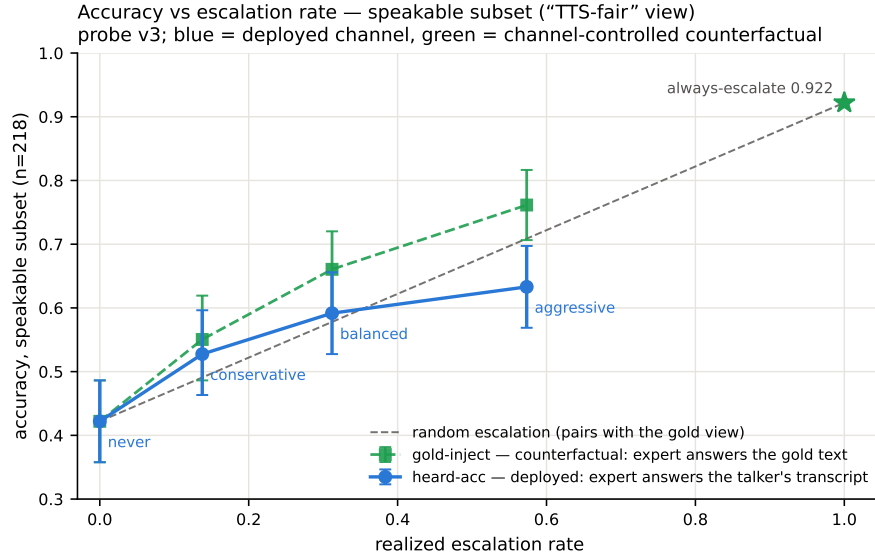


Figure 11: Dual-view live tradeoff on the speakable subset ($n=218$). Blue: honest heard-accuracy of the live duplex system (paired bootstrap CIs). Green: the channel-controlled counterfactual — same live sessions, escalated rows re-scored with frozen gold-text expert answers; its random reference (dashed) uses the same gold outcomes, so it pairs with the green curve only. The two annotated gaps itemize the deployment price: floor-to-floor, the audio + streaming-answer tax; view-to-view, the speech-channel (ASR-distill + relay) cost. Full-pool version: **[TODO: appendix figure (live_dualview)]**.

was a *cost* bug, not an accuracy bug. The interesting content is the *decomposition* of the gap to the offline curve, which the two views separate:

- **Audio + streaming-answer tax (floor .588 → .383 full pool, .422 speakable).** The live floor stacks the audio modality tax (§6) with a streaming-answer tax: in duplex voice mode the model produces short, speech-styled answers that sometimes plan aloud and stop. This is real deployment behavior — it is exactly why the never-escalate endpoint had to be run live rather than synthesized from chat-mode replays.
- **Speech-channel cost (+.023[†]/.069/.128 on the speakable subset; +.175 at aggressive on the full pool).** On the escalated population, the expert answering the talker’s own transcript scores .585 vs. .871 on gold text ($n=147$ escalated ids). The un-speakable share of this loss is exactly what the fairness filter removes; the deployment-real residue is entity loss on hard-knowledge: prompt-level rescues on the self-transcript are no-ops (a recognition-errors warning changes nothing; k -best self-transcription trades a small trap gain for a larger knowledge loss) because the damaging errors are internally-coherent entity substitutions with no textual residue to correct from. Replacing the *ears* is different from repairing the transcript: open-weights whisper-large-v3 recovers little (+4 pp), but routing the same wav through a stronger cloud ASR (gpt-transcribe) and handing *that* text to the expert scores .694 vs. .585 (paired McNemar $p=.007$), recovering 38% of the gap to the gold ceiling — a diagnostic result for now: its latency on the critical path and its survival on the external pools are unmeasured **[TODO: add scope note where input-side limits are stated]**. A trace-level decomposition of the escalated heard-failures (first-generation sweep; **[TODO: re-run on v3 traces]**) confirms the loss sits upstream of the relay: 64% are corrupted transcripts (the expert answers the wrong question), 24% are expert misses on clean transcripts (dominated by trap — an expert knowledge limit shared with the gold arm), and only 5 cases show a correct expert answer dropped in relay — consistent with the measured relay compliance of 1.00 on the deployed channel and with expert micro-turn

Table 9: Measured response latency (end of user query $\rightarrow$ answer text complete; seconds), v3 live sweep traces, $n = 240$ per arm (latency is channel-independent, so the full pool is used). Δ columns: loss vs. the never floor at the same statistic. Text-mode pipeline; speech synthesis excluded. P99 on $n = 240$ is indicative only.

tier	mean	P50	P95	P99	Δ mean	Δ P50	Δ P95	Δ P99
never (floor)	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	−3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

answers being short (median 76 characters), which rules out talker context length as a mechanism. A final controlled arm closes the attribution: sending the *original wavs* directly to an audio-capable expert (gpt-audio), with a same-model gold-text control to subtract its capability gap, shows the audio channel itself is lossless for speakable content — trap and easy-fact lose zero crossing it, and the same recordings the talker garbled (“999 — 103” with the minus dropped; “Mustafa Adebayo Balogun”) are heard verbatim by the stronger ear. The synthesis is therefore exonerated along with the relay; the leak is the talker’s own transcription step. We treat ASR robustness as an orthogonal channel property, not a contribution of this system.

The replication-noise floor. Live arm accuracies on a 200–250-query pool carry substantial judge-and-decode replication noise, and we report it rather than narrate inside it: the same query, same audio, kept *local* in two different arms flips its judge verdict 2.3–18.8% of the time depending on pool (this pool: .155); repeated *escalation* flips 0.7–10.6%. The implied paired SE on an arm-vs-arm accuracy delta is .009–.028, so any live delta under ~ 3 points is undetectable in a single sweep — accordingly, no claim in this paper rests on a sub-3-point live difference; probe comparisons rest on offline AUC and matched-rate precision, which are far tighter statistics on the full score distribution. **[TODO: noise_audit methods figure — appendix]**

What escalation costs in wall-clock time. The same frozen traces carry per-session timers, so the latency price of each operating point is measured, not modeled (Table 9). The gate itself is free at this scale: the end-of-turn read and the stall prefill cost ~ 30 ms each at P50; the expert round-trip (GPT-5.5, low effort; P50 4.8 s, P95 20.8 s) dominates escalated-session wall time (escalated-session P50 5.1–7.5 s vs. local 1.9–3.0 s). The consequence is a specific distributional signature: at the balanced tier the *mean* response time rises only +1.2 s and the P50 +1.5 s over the never floor, but the P99 nearly doubles (17.8 $\rightarrow$ 32.7 s) — escalation is cheap on average and expensive exactly in the expert-latency tail ($n = 240$ per arm, so P99 rests on the two worst sessions and is indicative only). The conservative tier even *shortens* the tail (P95 15.8 $\rightarrow$ 12.8 s): its few escalations displace exactly the longest local decodes, the same median-of-a-mixture mechanism analyzed on the external pools (§9). Figure 12 replays three real balanced-arm sessions from their recorded timers, showing the three shapes a session takes: the expert beating the stall phrase (seamless handoff), the expert outliving it (dead air — the tail case above), and the gate staying closed (first token in 0.4 s). Figure 13 joins the accuracy and latency readouts into the exchange rate actually paid at the median (speakable subset): +10.6 points of heard accuracy for +0.3 s (never $\rightarrow$ conservative), +6.4 for +1.0 s (conservative $\rightarrow$ balanced), and +4.1 for +1.1 s (balanced $\rightarrow$ aggressive). The first transition is both the steepest and the cheapest per second, for two measured reasons: escalation displaces local decodes that average 4.6 s (and the conservative tier’s escalations are precisely the slowest of them, hence its P95 shortening above), while the marginal escalations added at higher tiers reach easier queries whose experts return faster (escalated-session P50 7.5/6.9/5.1 s across the three tiers).

Three live sessions, timestamped (balanced arm; every event time is a recorded live-loop timer)

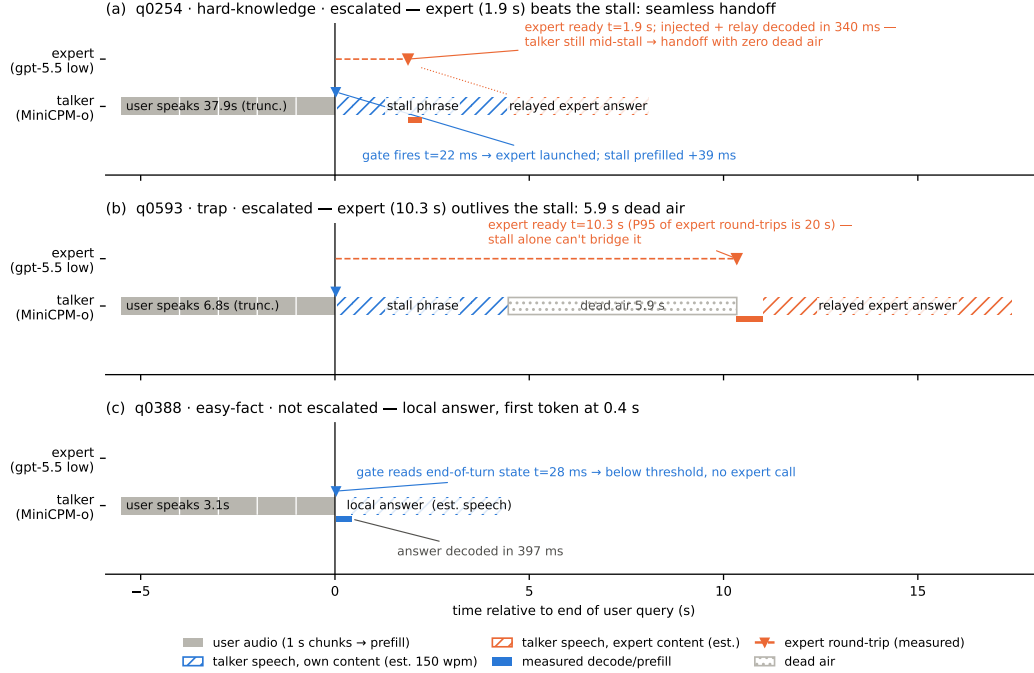


Figure 12: Three real live sessions (balanced arm), replotted from their recorded timers. (a) The expert returns at 1.9 s, while the talker is still voicing the stall phrase: the answer is injected mid-stall and the handoff has zero dead air. (b) The trap-pool case: the expert takes 10.3 s, the stall ends first, and 5.9 s of dead air follow — the P99 signature of Table 9 as a single session. (c) The gate stays closed and the local answer’s first token arrives at 0.4 s. Hatched speech bars are the one estimated quantity (150 wpm; the live loop emits text); every event timestamp is measured.

8.4 Signal boundary: a third blind spot

A fairness challenge to the trap pool — SimpleQA entities are hard to *hear*, conflating knowledge difficulty with transcription difficulty — motivated auditioning a transcription-fair failure pool: false-premise questions [Hu et al., 2023], every word common by design ($n = 60$, criterion fixed before data selection). The audition fails *as a pool* and succeeds as a finding: the small model fails substantially (.63 text / .47 audio) and the expert would help (.80 adequacy), but the end-of-turn gate is **blind** to these failures — fire@balanced .18 (vs. trap’s .90), with essentially no score separation between failed and passed answers (.568 vs. .507). A model answering along a false premise does not “feel” unable: the mid-layer signal reads *missing knowledge*, not *broken question*. This is the third documented blind spot, after easy-chat failures and decode-time math: the pre-answer signal is specifically a *retrieval-failure* detector. FalseQA therefore enters the signal-boundary analysis, not the tradeoff pools.

A boundary case that data can move: real-time queries. A fourth query class surfaced in live demo use: questions whose answer is unknowable at any training cutoff (“what is NVDA trading at right now”). Unlike false premises, the gate is not blind here — v3 scores sit *on* the balanced threshold, flipping with the TTS render (.624 vs. .693 for the same question) — the class was simply absent from every (static-knowledge) training family. Extending the calibration pool with FreshQA [Vu et al., 2023] closes it: fast-changing questions (153, labeled escalate a priori — the current value is unknowable by construction, so a judge against a stale gold would only add noise) plus never-changing in-family controls (150, standard judge labels, fail rate .47). Refit with the frozen v3 feature config, the v4 probe fires on held-out fast-changing questions at .80/.93/1.0 across tiers

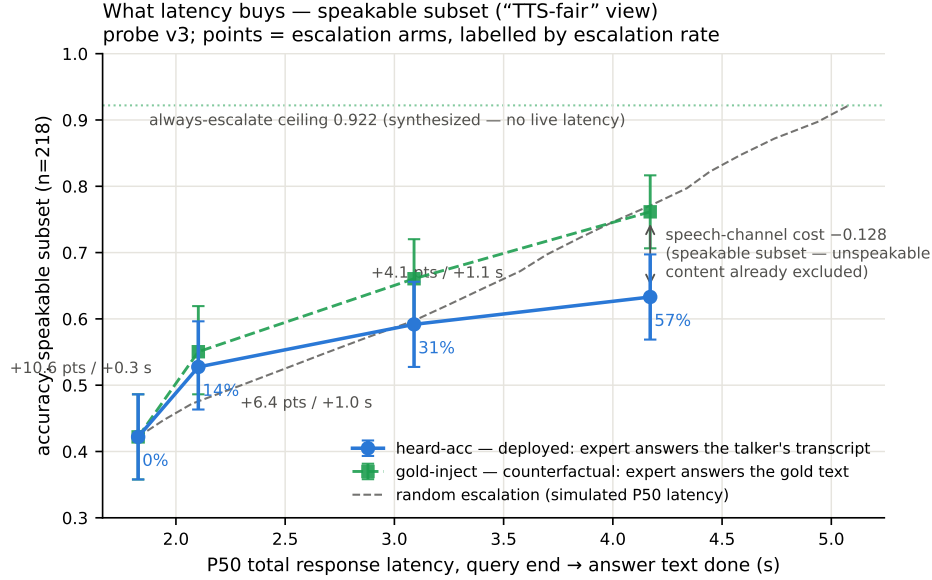


Figure 13: The live accuracy–latency tradeoff on the speakable subset: per-arm P50 response latency against accuracy with 95% paired-bootstrap CIs (Table 8), in the honest heard view (blue) and the channel-controlled gold-inject view (green) of the same sessions — rescoring changes outcomes, not timing, so the two views share their x positions. Segment labels give the median exchange rate on the heard curve; the dotted curve is the simulated-P50 random-escalation reference. The always-escalate ceiling is synthesized from frozen gold answers and has no live latency. Median view — the expert P99 tail is priced in Table 9. Full-pool version: [TODO: appendix (pareto_latency)].

(v3: .57/.80/1.0) while never-changing held-out rates and frozen-test AUC are unchanged (.877 vs. .879). One calibration subtlety: tier budgets must remain quantiles of the *deployment-mix* scores; quantiling over the augmented train mix (123 extra all-positive rows) silently raised the balanced threshold from .680 to .737, spending the new capability on its own threshold shift. Escalation only pays for this class if the expert can reach live data — the demo’s expert therefore gains a web-search tool, while all measured arms keep the tool-free expert.

9 Transfer Without Training: The Frozen Gate on Six Public Speech Benchmarks

Everything so far is measured on our internal pool, and §4.2 showed exactly why that is not enough: in a mixed pool, most of a router’s apparent margin can be the pool/type shortcut. This section is the out-of-distribution test the audit demands. We take the gate *frozen* — probe weights, feature configuration, and layer choice fit once on the internal calibration mix — and run the complete live streaming loop of §8 on six public speech-QA benchmarks: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [TODO: cite: OpenAudioBench; Web Questions (Berant et al. 2013); Llama Questions (Nachmani et al.)] [Joshi et al., 2017], SD-QA (real dialectal human speech, not TTS) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All pools use the benchmarks’ official pre-rendered audio — none of it passes through our TTS — and none of their queries ever touches probe training. The deployed probe (v3) is the same L22 end-of-turn read plus two additional zero-latency positions (a rolling 8-token tail and the running user-audio mean), refit on a calibration mix widened to 2,310 queries drawn from public families (SVAMP, HotpotQA, PopQA, MMLU, TruthfulQA-style traps) that are disjoint from — and deduplicated against — the six evaluation benchmarks; the pre-registered guard held with headroom (frozen internal test AUC .860→.879, and feature selection never saw the externals). The only per-pool adaptation is *label-free*: escalation-tier thresholds are

Table 10: The frozen gate on six public speech benchmarks, live streaming loop, v3 probe, realized escalation rates 15/30/50%. Judge: OAB = the benchmark’s official OpenAudioBench judge; VB = VoiceBench’s official 1–5 judge; ours = reference-anchored (no official protocol exists). Ceiling = GPT-5.5 on gold question text under the same judge; AUC = v3 probe score vs. never-arm local-fail label; official = MiniCPM-o 4.5’s reported number (offline chat mode; our own chat-mode control reproduces it, see text). Per-arm live accuracies carry a replication-noise floor of $\pm .02-.03$ (§9.3).

pool	judge	n	escalation tier				ceiling	AUC	official
			0%	15%	30%	50%			
Speech TriviaQA	OAB	250	.664	.732	.800	.860	.972	.789	.755
Speech Web Questions	OAB	250	.57	.628	.680	.732	.844	.785	.702
Llama Questions	OAB	250	.84	.904	.904	.948	.928	.806	—
SD-QA	ours	200	.51	.600	.720	.795	.930	.792	—
Reasoning QA (zh)	ours	202	.59	.653	.713	.772	.871	.683	—
AlpacaEval (1–5)	VB	199	3.99	3.96	4.23	4.35	4.96	—	4.8

quantiles of the probe’s own score distribution on each pool, a normalization the score scale forces (the aggressive threshold is .111 on Llama Questions vs. .603 on SD-QA — a $5\times$ shift for the same probe) and one that consumes no labels, no answers, and no judge calls.

Scoring is each benchmark’s own where a protocol exists: the three OpenAudioBench pools use the official judge verbatim (alias-match grading), AlpacaEval uses VoiceBench’s official 1–5 judge, and SD-QA / Reasoning QA (which ship no replicable protocol) use our reference-anchored judge. Under the official judge our offline chat-mode control *reproduces* the anchors — .712 vs. the official .755 on Speech TriviaQA (250-item subsample + protocol) and .716 vs. .702 on Web Questions — so the official lines in our figures are on-scale, and the live floor’s remaining gap to them is fully accounted (striviaqa: $-.048$ streaming-loop tax, $-.043$ subsample/protocol; nothing unexplained).

9.1 Accuracy: the mechanism ports, and the anchors fall

Table 10 is the headline: the frozen gate transfers mechanically everywhere — thresholds fire at sane rates, every curve rises monotonically, and the 50%-escalation arm lifts the live floor by $+.11$ to $+.29$ across the five QA pools. Three anchor results:

Routing beats scaling the speech backbone. On Speech TriviaQA the 9B duplex model plus the frozen gate at 50% escalation scores **.860 live**, against .755 for MiniCPM-o 4.5’s own official offline number and **.629** for Qwen3-Omni-30B-A3B ($3.3\times$ the parameters, offline chat mode) [TODO: cite: Qwen3-Omni / Kimi-Audio comparison rows (official OpenAudioBench table; Kimi-Audio scores .419 here)]. On Web Questions the gated system clears the official number (.732 vs. .702) and lands just under Qwen3-Omni (.749).

On Llama Questions, selective escalation matches the expert at half the cost. The pool is single-type (short English factoids), so no type shortcut exists to exploit, and the local model is strong (floor .84). The probe still splits it cleanly: at 50% the kept-local half scores **.976** vs. **.696** on the escalated half ($z=6.46$) — genuine per-query discrimination — and the expert’s entire advantage concentrates where the probe sends it (escalated half .696 $\rightarrow$.888, McNemar $p<.0001$) while buying nothing measurable on the kept half (.976 vs. .968, $p=1.00$). Selective at 50% reaches .948 vs. .928 for always-escalate (directionally above, $p=.125$ — both near ceiling): escalating everything spends twice the expert calls to buy nothing measurable over the gate.

Router quality must be read as AUC plus matched-rate precision, not raw precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail .16) the cap at 50% is .32, and the v3 gate’s matched-rate precision of .304 is **95% of the theoretical maximum** (recall .93); quoting “.30 precision” without the cap would misread the strongest pool as the weakest. The scale-free readout is transfer AUC against the never-arm fail label: .68–.81 on every external pool (Table 10) vs. .879 on the internal held-out test — the mid-layer signal ports as a generic difficulty signal, with a near-uniform matched-rate precision lift of $\times 1.33$ – 1.35 across pools rather than pool-

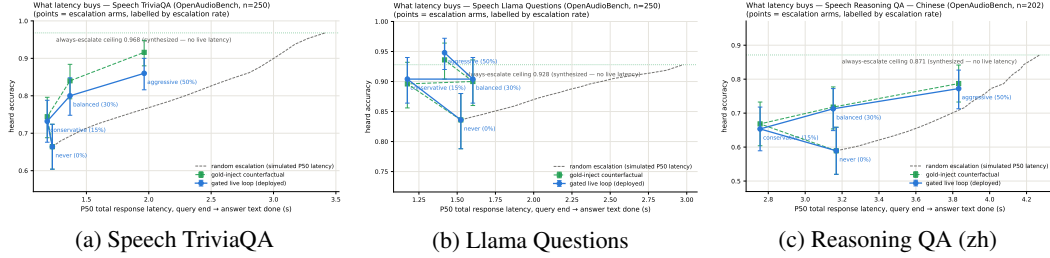


Figure 14: Accuracy vs. per-arm P50 response latency on three external pools (frozen gate, live arms; official judge on a–b, ours on c). The random reference simulates a random mixture at each rate from per-id *measured* latencies (400 sims $\times$ 21 rates). On (a) and (b) the gated curve dominates random in *both* axes: on Speech TriviaQA random needs ~ 3.4 s at the median to reach the accuracy region the gate reaches at 2.0 s, and on Llama Questions random at 50% sits at ~ 2.2 s/.88 vs. the gate’s 1.42 s/.948. On (c) escalation truncates the latency *tail* (P90 13.25 $\rightarrow$ 10.94 s). Remaining family figures: **[TODO: appendix: dualview + pareto for all six families]**.

specific structure. The cleanest evidence that the signal is difficulty, not surface, is cross-lingual: the calibration expansion is *English-only*, yet it moves Reasoning QA — a *Chinese* execution-reasoning pool — from AUC .621 to **.683** (+.062), the largest per-pool gain of the refit. Per the replication-noise analysis below, the v3 evidence is this offline AUC shift; the corresponding live-arm deltas (+.010–.030) are individually within noise and we do not narrate them.

9.2 Latency: the probe escalates the slowest decodes first

The latency curves of Figure 14 have a shape worth explaining honestly, because at first sight it looks like a measurement error: on the easy pools the per-arm P50 is *non-monotonic* — on Llama Questions it *falls* from 1.52 s (never) to 1.17 s at 15% escalation before rising again (1.60/1.42 s at 30/50%). This left-fold is a verified median-of-a-mixture mechanism, not noise: the probe preferentially escalates exactly the queries whose *local* decode is longest. On Speech TriviaQA the 38 queries escalated at 15% had a never-arm local P50 of 1.711 s vs. 1.198 s for the 212 kept — and, measured in the never arm, local accuracy .32 vs. .73: the small model’s signature failure mode on retrieval pools is hedging at length, so wrong answers are slow answers (decode time and answer length are the same variable, $r=.97$), and a wrongness-selecting probe strips the slow tail of the local mixture as a side effect. Controls: the probe does not key on length directly ($\text{corr}(\text{score}, \text{length}) = +.05$; conditioned on wrongness the score–length coefficient is ≈ 0), and on SD-QA — where wrong answers happen to be short and confident — the same probe selects wrongness identically and produces *no* fold. Figure 15 decomposes the effect: the fold lives only in the median — the *mean* is monotonic (1.65/1.81/1.96/2.06 s vs. P50 1.52/1.17/1.60/1.42 s) — and the kept-local P50 falls 1.52 $\rightarrow$ 0.96 $\rightarrow$ 0.86 $\rightarrow$ 0.43 s as the probe strips slow decodes while escalated rows pay a flat ~ 3 s expert round-trip. We report the mean alongside the median for exactly this reason; the honest statement is that on these pools escalation buys accuracy *and*, at the median, speed. The individual dip depth is within noise (striviaqa -41 ms, bootstrap CI $[-285, +254]$ ms) but its sign is deterministic — identical probe scores select the identical escalated set on every rerun.

On Reasoning QA the shape inverts usefully: P50 rises gently but the *tail improves* — P90 drops from 13.25 s to 10.94 s under escalation. The mechanism is that the local model’s chain-of-thought is *spoken*, so every reasoning token enters the response clock (escalated-set local decodes P50 4.38 s, tail 13 s+), while the expert’s chain is hidden: its median completion is 144 tokens of which only 57 characters return, behind a flat ~ 3.7 s round-trip. On reasoning pools the gate replaces the longest spoken chains with short round-trips — escalation truncates the latency tail rather than fattening it.

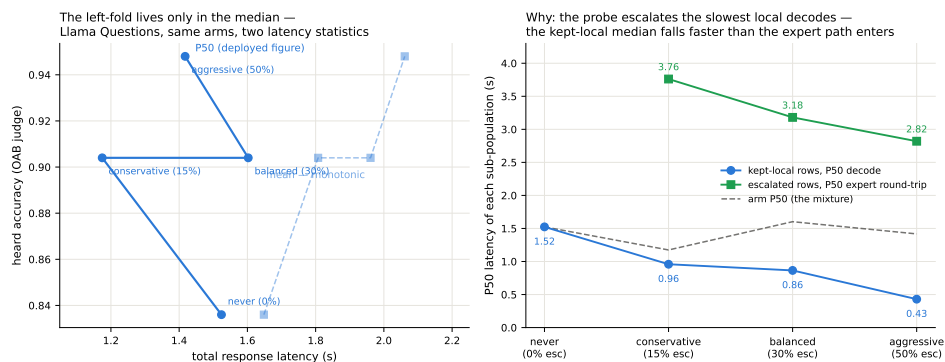


Figure 15: Decomposition of the Llama Questions latency fold. Left: the fold exists only in the median — the mean is monotonic. Right: kept-local P50 falls 1.52→0.43 s across tiers as the probe strips the slowest local decodes, while escalated rows pay a flat ~3 s expert round-trip; the arm median folds left and then rises as the expert mass reaches it.

9.3 Where transfer fails, and what the live curves can and cannot show

AlpacaEval is a negative result, mechanism measured. On VoiceBench AlpacaEval the gate does *not* beat random: aggressive 4.35 vs. random-at-50% ≈ 4.45 . The failure is selection, not escalation — queries the gate selected score 3.90 on the never arm vs. 3.98 for those it skipped (no discrimination), while escalated rows still gain (3.90→4.71) purely because GPT-5.5 writes better long-form answers; random picks would harvest the same gain. Open-ended instruction following has no “the model lacks this fact” event for a retrieval-failure detector to read, and the score range is compressed (1.0 point vs. 40 accuracy points on our pool). Two scale caveats make the absolute level honest: the official 4.8 is an *offline chat-mode* number — our chat-mode control reproduces it at 4.86 — and the entire chat-vs-live gap (4.86 vs. 3.99) is the duplex loop’s spoken-reply style plus a 512-token cap, so only the live arms are comparable to each other.

Web Questions and SD-QA are flat under the v3 refit, offline (AUC $\pm .006/\pm .017$) and live alike. On Web Questions a specific share of the blame is label fidelity rather than the model: its strict Freebase-alias grading decouples “wrong” from “uncertain” — the wrong-answers-are-longer signature is null there for every model we measured, including GPT-5.5 on gold text — so a portion of its labels is protocol-wrong that no state-reading probe can predict, and even the gold-text expert ceiling is .844.

The deployed view understates selection on easy pools. The external channel cost is small — the aggressive-arm heard-vs-gold-inject gap is .040/.024/.025 on striviaqa/swebq/sdqa [TODO: verify: gold-inject gaps on the v3 traces; quoted values are from the v2-probe sweep] vs. .175 on the internal pool (.128 on its speakable subset, §8.3), exactly as predicted for short open questions with no option walls — but on easy homogeneous pools even a small tax eats a thin per-query margin, and part of the deployed-view “gate $\approx$ random” on Speech TriviaQA is channel cost: in the channel-controlled gold view the gate clears its paired random line (+.044 at the aggressive arm). SD-QA meanwhile carries the one property no TTS pool has — real dialectal human speech — and the escalated subset still converts at .82–.84 with expert P50 3.1–3.4 s: the payload survives real speech.

Replication-noise discipline. Re-running the same query with the same audio through the live loop flips the judge verdict 2.3–18.8% of the time depending on pool (Llama Questions .023, SD-QA .188), implying a paired SE of .009–.028 on any arm-vs-arm accuracy delta. Against that floor, all 18 per-arm live v2→v3 deltas are individually non-significant (the largest, striviaqa balanced +.036, gives McNemar $p=.16$). We therefore treat sub-3-point live deltas as unresolvable in a single sweep and rest every probe-improvement claim in this section on the variance-reduced instruments — offline AUC over the full score distribution and matched-rate precision — with the live curves serving as the end-to-end existence proof, not the measurement.

10 Discussion

10.1 Design consequences: the two-stage gate

The evidence converges on a two-stage design:

- **Stage 1 (prefill, effectively free):** a mid-layer ($\sim 60\%$ depth) last-token probe — 20/45 ms, before the first token, and the only signal that survives duplex fine-tuning *and* both input modalities (§6.5). Open engineering item: score-scale calibration for threshold (rate) transfer.
- **Stage 2 (optional, post-draft):** $p(\text{True})$ -post on the drafted answer — the strongest single signal (0.899); the draft doubles as the distilled escalation query and as the fallback answer. On audio input, never ask over raw audio context: run $p(\text{True})$ on the model’s own transcript (“repeat-then-judge,” §6.3).

10.2 Positioning vs. query-side routing

Query-side routers [Ong et al., 2024, Ding et al., 2024, Chen et al., 2023] decide before generation from query features, and the big model then answers the user directly. Neither transfers to full-duplex voice: the user talks to one voice whose session state lives in the small model, and query-side features are precisely the “type shortcut” our audit quantifies (pool oracle 0.715 vs. probe 0.822 in-distribution) and LOPO breaks. The signal must come from the answering model itself — consistent with LLMRouterBench authors [2026]’s attribution of router failures to model-recall gaps. A head-to-head comparison of query-feature routers vs. internal-signal gates under distribution shift, on their standardized protocol, is natural future work.

10.3 Result injection: resolved by the same signal

Step 2 — returning the thinker’s result into the live duplex session — initially looked like a research problem of its own: the streaming smoke test showed the talker pushing back on conflicting injected results. The quantified answer (§8.2) is that for *gated* escalations the problem largely dissolves, and for a principled reason: the same low-confidence state that fires the gate predicts relay compliance (comply .42 $\rightarrow$.75 across gate-score quartiles; 1.00 on the deployed wrong-model $\times$ correct-expert workload). The gate and the relay are two uses of one signal. Where the state is confident, framing — not ventriloquism — is the mechanism (authority framing .53–.79 vs. seeding’s backfire), and the residual ceiling is the swallow rate: an unconfident talker relays a *wrong* expert at .64 with zero pushback, so cascade accuracy is bounded by expert quality.

10.4 Three failure species, one perceivability axis

The accumulated blind-spot results organize into a taxonomy of *why* a model fails, ordered by whether the failure is an observable event in the model’s own processing. **Retrieval-type** failures (“I don’t know this entity”) produce an observable empty lookup at prefill — pre-answer signals work (trap fire .90). **Execution-type** failures (“stuck mid-solution”) only come into existence during decoding — decode-time signals see them (math entropy@16 .860) and pre-answer signals cannot. **Metacognitive-type** failures (“doesn’t know the question itself is broken”) never occur as an event anywhere in the model’s experience — like a schoolkid confidently “verifying” Goldbach’s conjecture, the model answering a false-premise question executes smoothly along the wrong frame, and every internal signal is blind (FalseQA fire .18, no score separation; §8.4). This mirrors the human feeling-of-knowing literature, which finds metacognition accurate for memory retrieval and unreliable for insight problems. The taxonomy explains the two-stage gate architecture (§10.1 rationale: prefill probe for species 1, decode monitor for species 2), scopes the claim — the pre-answer signal is a retrieval-failure detector, not a general oracle — and notes that query-surface routers miss species 3 as well.

10.5 Demonstration system

The measured pipeline runs as an interactive web demo: live microphone audio streams to the duplex model on a single H100, the end-of-turn probe read costs ~ 54 ms against the balanced threshold, and escalated turns stall, consult the expert, and *speak* the relayed answer through the model’s own TTS head. Two demo-only extensions go beyond the measured arms: the expert gains a web-search tool — real-time queries such as stock prices, which the v4 FreshQA refit (§8.4) fires on at .80–1.0 across tiers, then return live values no cutoff-bound model can produce — and the loop accepts *backchannel-aware barge-in*: sustained speech over the talker ducks playback, a ~ 1 s ASR pass classifies the burst, and backchannels (“ok”, “mm-hm”; an English+Chinese lexicon) return the floor with zero loss while anything else aborts generation and seeds the next turn. All measured arms in this paper keep the tool-free expert and the turn-based loop; the demo will be made available upon publication.

10.6 Limitations

(i) $p(\text{True})$ prompt sensitivity: one phrasing each (pre/post). (ii) The audio arm is validated on real dialectal speech (SD-QA), but scripted read speech and one dialect split; spontaneous conversational speech (disfluencies, self-repair) remains uncovered. (iii) Mid-layer probe threshold transfer needs deployment-grade calibration; AUC/area claims are unaffected. (iv) The audio side of the introspection-collapse finding is MiniCPM-scoped (two generations of one family); a new open-weight full-duplex family is a pre-registered prediction test — it should show the late-layer text-input cliff. (v) The frozen test split is reused across signal comparisons (signals were pre-specified; a fresh test set would be cleaner). (vi) Labels come from a single judge family, bounded by a stronger re-judge ($\text{deltas} \leq 0.010$) but without human labels. (vii) The live sweep (§8) is one session per query per tier; the conservative tier’s gain over the floor is not significant at $n = 240$. The measured live sweep ran with text-mode talker output (`generate_audio=False`): the relay is validated at the utterance level, and its latencies are text-complete — a conservative accounting, since in the spoken system the stall filler plays the moment the gate fires. The released demo now speaks with the model’s native TTS head (talker + Token2wav vocoder, 24 kHz; first answer audio ~ 0.9 –1.4 s after generation start; the TTS template token enters after the end-of-turn read, so the probe context and all frozen thresholds are unchanged). The demo also supports barge-in with backchannel awareness — speech over the talker first ducks playback, a fast ASR pass then either returns the floor (“ok”, “em”) or commits the interrupt (“stop”, new content), which aborts generation at chunk granularity and seeds the next turn. This is a lexicon-level approximation of what a native duplex listen/speak head decides from hidden state; that head remains unused, since the probe is calibrated on simplex streaming states, and its validity under the true duplex loop is an open experiment.

11 Conclusion

Zero-training self-knowledge in a small full-duplex model exists, is strong, and is modality- and layer-fragile in systematic, duplex-fine-tune-induced ways. The standard readouts — final-layer probes and verbalized self-evaluation — each have one blind modality, and both blind spots are created by the duplex fine-tune; the mid-layer representation underneath survives everything we tested. Read there, the resulting gate decides before the first output token, transfers from cheap text calibration to speech on end-to-end models, and as a system dominates the model’s own thinking mode on both accuracy and latency. And when the thinker’s answer comes back, the same signal that fired the escalation predicts that the talker will actually say it: run live end-to-end, the gated duplex system climbs monotonically from its honest streaming floor (.422 $\rightarrow$.633 at 57% escalation on the speakable subset), with the remaining gap to the offline curve itemized into named, measured taxes rather than left as mystery — and the frozen gate transfers unchanged to six public speech benchmarks, where at 50% escalation it lifts the 9B system past models $3\times$ its size. Gate and relay are one mechanism; what remains is engineering the channel around it.

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TODO / Roadmap (working section — delete before submission)

[W] Must-write — ALL EXECUTED 2026-08-25 (peer-draft session)

- **[TODO: W1 — BIGGEST GAP:** the external-benchmark transferability section does not exist in the paper. live.tex has ZERO mention of striviaqa / swabq / sdqa / sllama / sreason / valpaca; none of the 12 dualview/pareto family figures are included. This was the 2026-08-12 meeting deliverable. Sources: RESULTS §8x (bench infra + latency shapes), §8z + 8z-live (v3 gains: striviaqa balanced .764→.800, sreason cross-lingual, sllama selective@50% .948), §8ab (framing rules). Anchors per figure: official MiniCPM-o 4.5 number (floor, 75.5 striviaqa / 70.2 swabq), GPT-5.5-on-transcript (ceiling), random escalation at matched rate (control). Framing rules from the Jisen pass (8ab): (a) cite that the gate dominates the simulated-P50 random curve in BOTH axes on striviaqa/sllama; (b) the P50 left-fold is a verified median-of-a-mixture mechanism — use sllama_latency_decomp, report mean alongside median; (c) frame router quality via AUC + matched-rate precision (v3 hits 95% of the base-fail/rate cap on sllama), never raw precision; latency-shape note: non-monotonic per-arm P50 on easy pools, tail truncation on Reasoning QA (P90 13.3→10.9 s)]
- **[TODO: W2 — 8ad noise audit: THREE MANDATORY REWRITES of §8,** still absent from live.tex (verified 2026-08-25: Table 8 still carries the v1 numbers .400/.446/.529/.633). (a) re-read every arm number off live_v3_figures.py output: heard .383/.483/.554/.596 at 0/16/35/61%, channel cost −.175, P50 2.02/2.61/3.53/4.44 s, P99 32.7 s at balanced. (b) sllama headline: “MATCHES the expert at half the cost” (.976 vs .968 McNemar $p=1.00$); what is significant is the probe’s split ($z=6.46$) and the hard-half lift .696→.888 ($p<.0001$); .948-vs-.928 is $p=.125$. (c) report the replication-noise floor (same-query flip 2.3–18.8%, paired SE .009–.028), stop narrating sub-3-point live deltas (all 18 v2→v3 arm deltas n.s.; v3 evidence is the offline AUC). Add noise_audit as a methods figure]
- **[TODO: W3 — second duplex family (8ac, EXECUTED 2026-08-19, not in any section):** NemotronLabs-VoiceChat-11B (hybrid Mamba), the pre-registered prediction test — mid-band layer structure replicates (L30–34 of 56), frozen-methodology transfer AUC .70–.79 with 600-query calib; sreason boundary (model is English-only). One paragraph in duplex.tex or discussion: this is the generality evidence for “mid-network read” beyond MiniCPM]
- **[TODO: W4 — cloud-ASR uplink (8ae, not in any section):** on 147 escalated frozen ids, gpt-transcribe → gpt-5.5 scores .694 vs .585 deployed self-transcript (McNemar $p=.007$), recovering 38% of the gap to the .871 gold ceiling; SUPERSEDES the “Whisper ears +4pp only” write-off (8d used whisper-large-v3). Write as diagnostic finding with both caveats (ASR latency on critical path unmeasured; external-pool survival unmeasured). hard-math recovers nothing — spoken LaTeX is an input-side limit, state it separately where input-side limits are stated]
- **[TODO: W5 — small syncs:** (a) deployment-artifact sentence: label-free quantile correction (gate_v3_thresholds_corrected.json) removes 11.3% of expert calls at equal accuracy on the frozen pool; externals already on-budget. (b) decide whether the demo system (8aj–8am: TTS head, v4 real-time gate, web-search expert, backchannel-aware barge-in) gets a short system/demo paragraph or an appendix; it is currently only in RESULTS. (c) FreshQA/v4 paragraph already in live.tex — verify numbers match RESULTS 8ak final (.877 frozen-test, fire .80/.93/1.0)]

[D] Decisions — ALL RATIFIED 2026-08-25 (user approved all three)

- **[TODO: D1 — speakable-subset headline swap (8q, user to ratify):** fairness audit removes 22/240 ids (21 LaTeX + 1 broken wav; criteria decidable from query text + wav bytes alone) ⇒ aggressive heard-acc .633→.688, channel gap .133→.083 (still

significant), gold ceiling unmoved. If ratified: dual-view + Pareto redrawn on the fair subset (fair_* figures already built), full-pool to appendix, criteria stated as pre-registered filters, update the 8m 64%-corrupted sentence in live.tex. NOTE: interacts with W2 — do the v1→v3 number swap and this in one pass]

- [TODO: D2 — DONE (2026-08-25, RESULTS §8an, scripts/07_bootstrap_oracle.py): point estimates reproduced (+.0541/+0.0422/+0.0119); paired bootstrap $B=10^4$ gives 95% CI $[-.003, +.027]$, $p=.13$ — NOT significant, as the 7-29 entry suspected. system.tex:21 softened to “statistically on par with the pool oracle in distribution”. No other untested-delta claim of this class remains]
- [TODO: D3 — double-blind exposure: the demo URL (rhe9527-gate-demo-web.modal.run) contains the author username; author block on the title page must go for submission (camera-ready restores it). Decide: anonymized demo link (new modal workspace / redacted) vs. “demo available upon publication”]

1013 [M] Mechanical

- [TODO: M1 — DONE 2026-08-25: neurips_2026.sty submission mode, author block anonymized, compiles under tectonic. Remaining: the 8-page main-text compression (currently far over)]
- [TODO: M2 — figure hygiene: (a) regenerate concept_probe_vs_ppl with English labels; (b) ROC figure to include $p(\text{True})$ curves; (c) verify pareto_latency rebuild (2026-08-20) actually contains the random curve (8ab-d); (d) vintage audit — any figure included in the paper with an artifact predating 2026-08-18 must be checked against v3 traces (currently included: concept_probe_vs_ppl, fork_pareto, layer_sweep, live_dualview, overlap, pareto_latency, receipt_compare, roc, timeline_live, tradeoff, tradeoff_midlayer, tradeoff_pttrue — the July-vintage ones are suspect)]
- [TODO: M3 — bib: verify LLMRouterBench authors [2026] (arXiv:2601.07206) and the MiniCPM-o 4.5 model-card entry]
- [TODO: M4 — surface existing bootstrap CIs in tables beyond Table 2]
- [TODO: M5 — appendix assembly: full per-pool tables, judge prompts, $p(\text{True})$ prompts, reproducibility statement (seeds, pins, spend)]

1030 [F] Future work — fold into discussion, then delete

- [TODO: one paragraph covering: (a) 10 remaining SD-QA dialect splits (mechanical robustness table, ~\$100); (b) true duplex integration — ls-head + L22 probe validity under the duplex loop (the probe is calibrated on simplex streaming states); (c) Nemotron remaining sub-items (live 4-arm port, calib expansion, trained-checkpoint ablation); (d) audio-direct-to-expert live sweep (the one lever a single sweep can resolve; gold/heard gap $-.175$ is an order above the noise floor); (e) trained-gate upper-bound baseline (the only legitimate “real training”: a contrast, not a method change); (f) LLMRouterBench protocol comparison; (g) $p(\text{True})$ prompt-sensitivity sweep; (h) injection payload knee (answer + one sentence justification vs. math relay compliance, \$5); (i) closed-source behavioral arm (GPT audio / Gemini Live on the 600 ways); (j) camera-ready: fresh test split + pool expansion to 500–1000/pool]